

SITE PREPARATION AND EARTHWORKS INCLUDING TANK FOUNDATIONS AND TANK FARMS

DEP 34.11.00.11-Gen.

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(DEP Circular 60/08 has been incorporated)

DESIGN AND ENGINEERING PRACTICE



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TABLE OF CONTENTS

1.	INTRODUCTION	5
1.1	SCOPE.....	5
1.2	DISTRIBUTION, INTENDED USE AND REGULATORY CONSIDERATIONS	5
1.3	DEFINITIONS	5
1.4	CROSS-REFERENCES	6
1.5	CHANGES FROM THE PREVIOUS EDITION	6
1.6	COMMENTS ON THIS DEP	7
2.	GENERAL	8
2.1	EXISTING SERVICES AND FACILITIES	8
2.2	SITE LEVELS.....	8
2.3	SITE INVESTIGATIONS	9
2.4	CLASSIFICATION OF SOILS AND ROCKS	9
2.5	SEISMIC ASPECTS	9
3.	SITE PREPARATION	10
3.1	GENERAL	10
3.2	DEMOLITION AND SITE CLEARANCE	10
3.3	STRIPPING OF TOPSOIL	11
3.4	REROUTING OF EXISTING DITCHES AND WATER COURSES	11
3.5	TREATMENT OF UNDERGROUND CAVITIES.....	11
3.6	TREATMENT OF SUBSOILS	11
3.7	PROVISION OF SITE DRAINAGE	11
3.8	PROVISION OF TEMPORARY ACCESS	12
3.9	LAY-DOWN AND STOCKPILE AREAS	12
3.10	SCOUR PROTECTION, COASTAL DEFENCE AND RIVER WORKS.....	12
4.	EARTHWORKS	14
4.1	GENERAL	14
4.2	CONTROL OF EARTHWORKS.....	14
4.3	PREPARATION OF SOIL BEARING FOUNDATIONS	15
4.4	MOUNDS FOR BURIED VESSELS	16
5.	BUNDED AREAS AND FOUNDATIONS FOR TANKS	17
5.1	BUNDED AREAS FOR TANKS	17
5.2	IMPOUNDING BASIN	17
5.3	BUNDS AND BUNDWALLS	18
5.4	SURFACE FINISHES OF BUNDED AREAS.....	20
5.5	TANK FOUNDATIONS	20
5.6	TANK FOUNDATION WITH LEAK AND DETECTION OF LEAK MANAGEMENT SYSTEM	28
5.7	SPECIAL TANK FOUNDATIONS	29
6.	TANK MAINTENANCE	31
7.	DRAINAGE OF BUNDED AREAS	33
8.	REFERENCES	34

APPENDICES

APPENDIX 1	FIELD COMPACTION TRIALS – SUGGESTED PROCEDURE.....	36
APPENDIX 2	INDICATIVE COMPACTION VALUES.....	37
APPENDIX 3	MIXTURES OF SAND AND BITUMEN FOR TANK FOUNDATIONS	40
APPENDIX 4	WET SAND MIX FOR TANK FOUNDATION SHOULDERS AND BUND WALLS.....	41
APPENDIX 5	TYPICAL CPT VALUES TO VERIFY COMPACTION AND QUALITY OF TANK FOUNDATIONS	42

APPENDIX 6	HYDROSTATIC TESTING PROCEDURE	44
APPENDIX 7	TYPICAL TANK FARM LAYOUTS WITH SAFETY DISTANCES	47
APPENDIX 8	TYPICAL EXAMPLES OF RETAINING WALLS	49
APPENDIX 9	EFFECT OF TANK DIMENSIONS ON SOIL SETTLEMENT	50
APPENDIX 10	TYPICAL RING FOUNDATION WITH CONCRETE SLAB	51
APPENDIX 11	SOIL IMPROVEMENT BY GROUND REPLACEMENT	52
APPENDIX 12	EXAMPLES OF TANK FOUNDATION CONSTRUCTION	54
APPENDIX 13	MATERIALS	57
APPENDIX 14	NUMBER OF VERIFICATION CPTs FOR COMPLETED TANK FOUNDATION	62
APPENDIX 15	TYPICAL MINIMUM CONE RESISTANCES FOR SOIL COMPACTION	63

1. INTRODUCTION

1.1 SCOPE

This DEP specifies requirements and gives recommendations for the design, planning, material selection, construction and maintenance for site preparation, earthworks, and shore and scour protection. It also covers the same elements for the civil works of tank farms and tank foundations.

This DEP is a revision of the DEP of the same number, previously titled Site Preparation and Earthworks and dated November 1983. It has been updated to incorporate the subject of tank foundations and bund walls, previously covered by DEP 34.18.51.10-Gen., and which is now withdrawn.

In preparation of this DEP, the prominent and relevant points from BS 6031 have been incorporated. In the event of doubt, reference can be made to that standard.

The information contained in this DEP is supplementary to the following civil engineering DEPs:

Minimum requirements for structural design and engineering	DEP 34.00.01.30-Gen.;
Site investigations	DEP 34.11.00.10-Gen.;
Geotechnical and foundation engineering	DEP 34.11.00.12-Gen.;
Roads, paving, surfacing, slope protection and fencing	DEP 34.13.20.31-Gen.;
Drainage and primary treatment facilities	DEP 34.14.20.31-Gen.

1.2 DISTRIBUTION, INTENDED USE AND REGULATORY CONSIDERATIONS

Unless otherwise authorised by Shell GSI and SIEP, the distribution of this DEP is confined to companies forming part of the Royal Dutch/Shell Group or managed by a Group company.

This DEP is intended for use in oil refineries, chemical plants, gas plants and exploration and production facilities as well as distribution storage depots and installations. When DEPs are applied, a Management of Change (MOC) process should be implemented. This is of particular importance when existing facilities are to be modified.

If national and/or local regulations exist in which some of the requirements may be more stringent than in this DEP the Contractor shall determine by careful scrutiny which of the requirements are the more stringent and which combination of requirements will be acceptable as regards safety, environmental, economic and legal aspects. In all cases, the Contractor shall inform the Principal of any deviation from the requirements of this DEP which is considered to be necessary in order to comply with national and/or local regulations. Based on the potential impact on design, construction and maintenance costs, the Principal may decide to consult with the Authorities concerned with the object of obtaining agreement to follow this DEP as closely as possible.

1.3 DEFINITIONS

1.3.1 General

For the purposes of this DEP the following definitions shall apply:

The **Contractor** is the party that carries out all or part of the design, engineering, procurement, installation, and commissioning or management of a project or operation of a facility. The Principal may sometimes undertake all or part of the duties of the Contractor.

The **Manufacturer** is the party that manufactures or supplies equipment and services to perform the duties specified by the Contractor.

The **Principal** is the party that initiates the project and ultimately pays for its design and construction. The Principal will generally specify the technical requirements. The Principal may also include an agent or consultant, authorised to act for the Principal.

The word **shall** indicates a requirement.

The word **should** indicates a recommendation.

1.3.2 Specific definitions

Environmental engineering - engineering to protect the environment of those who may be affected by the activities

Amended per
Circular 60/08

Very toxic (substances) - substances that are very hazardous for the environment or human health, as specified in DEP 01.00.01.30-Gen. (which also identifies "toxic" substances by reference to chemical substances databases).

1.3.3 Definitions of materials

For the definitions of materials, see BS 6100 – 1.0 and BS 6100 – 2.2.2.

1.4 CROSS-REFERENCES

Where cross-references to other parts of this DEP are made, the referenced section number is shown in brackets.

Other documents referenced by this DEP are listed in (8).

1.5 CHANGES FROM THE PREVIOUS EDITION

The previous edition of this DEP was dated November 1983. This revision updates DEP 34.11.00.11-Gen. and also incorporates the requirements and recommendations for tank foundations and bund walls previously covered in DEP 34.18.51.10-Gen. Other than changes involving section numbering, formatting, or minor editorial updates, the following are the main technical changes to these DEPs:

Old Section		New Section	Change
DEP 34.11.00.11	DEP 34.18.51.10		
2.3		2.3	Environmental Baseline survey added as integral part of site investigation.
2.5		2.5	Requirements added reflecting the increased general and public knowledge on the seismic hazard assessment subject.
3.2		3.2	Items added to highlight segregation and quantification of hazardous materials.
5		3.10	Environmental Impact Assessment added as part of the requirements and emphasis to economic aspects to be considered.
6	4,5,6	5	Detail increased, reflecting the general increase in knowledge, especially regarding geotechnical aspects and soil investigations. Emphasis added towards environmental impact, safety (fire) hazards and economic impact.

Old Section		New Section	Change
DEP 34.11.00.11	DEP 34.18.51.10		
	-	5.6	Section added on tank leakage control.
	-	Appendices 11 through 15	Geotechnical information added on ground improvement and foundations.

1.6 COMMENTS ON THIS DEP

Comments on this DEP may be sent to the DEP Administrator at standards@shell.com.

Shell staff may also post comments on this DEP on the Surface Global Network (SGN) under the Standards/DEP 34.11.00.11-Gen. folder. The DEP Administrator and DEP Author monitor these folders on a regular basis.

2. GENERAL

2.1 EXISTING SERVICES AND FACILITIES

Prior to beginning site investigations, site preparation works and earthworks, a certificate or permit shall be obtained from the Principal or the responsible body. Relevant utility companies and/or local authorities shall be contacted to establish whether they have services in the subject locations. This should normally be done by the Principal and shall confirm that services in the locality have been made safe, and any services or other facilities adjacent to the proposed work shall be described. In addition, any safety aspects and measures that are to be maintained during the work shall be specified. This shall include ensuring that all aboveground and underground services shall be maintained during the works through physical support if necessary.

The stability of all supports or excavations shall be proven by calculations based on realistic assumptions and/or available data. The nature and location of all existing aboveground and underground services, and facilities shall be determined, where necessary by hand excavation. Typically, underground services are located up to a depth of 1.50 m below ground level. Applicable excavation permits shall be adhered to for any site work.

Every precaution shall be taken, including the provision of all necessary temporary supporting, bridging, shoring and safety barriers to protect these services and facilities from damage or interference during the execution of the works. The stability of existing aboveground and underground services shall be maintained during the works in accordance with the requirements of the applicable permit or certificate to work.

If any previously unrecorded services or other objects are discovered during investigations, the Principal shall be notified in writing. An accurate description, including dimensions, levels related to reference datum and other relevant details of such services, shall be prepared and supplied to the Principal.

If services such as drains or natural drainage of a certain area have to be interrupted or otherwise interfered with during the execution of the works, these shall be made good to the satisfaction of the Principal. Where applicable, temporary or permanent rerouting shall be provided.

If services or facilities are accidentally cut or damaged, these shall be repaired as directed by the Principal.

2.2 SITE LEVELS

During the process of establishing finished grade levels, e.g., high point of paving, the following shall be taken into account:

- Surface water should be able to run off under gravity via the plant drainage systems, in accordance with DEP 34.14.20.31-Gen.
- Site layout/plot plan and related safety aspects, including safety distances, should be adhered to.
- Cut and fill should be economically planned so that cut and fill balancing is optimised, including allowances for earthen structures and general fill, bulking, wastage and unsuitable materials.
- Danger of flooding should be avoided (i.e. highest sea level, storm surge, wave run-up, highest river level, etc. should be determined) or alternatively the area should be adequately protected.

Prior to the commencement of work, a limited number of permanent monuments (benchmarks) shall be strategically installed on site and accurately related to the plant and/or national grid system for line and level.

2.3 SITE INVESTIGATIONS

Prior to the start of site preparation and earthworks, a site investigation shall be carried out. This investigation should include, as a minimum, soils, topographical, hydrological, hydrogeological, meteorological and rainfall intensity surveys. In certain instances, engineering geological surveys may be required.

As part of the site investigation, an environmental baseline survey is often carried out which typically comprises the installation of monitoring wells and the sampling and analysis of groundwater and soil samples to determine concentrations of contaminants. The installed monitoring wells allow an ongoing programme of sampling and testing to be carried out.

The design, materials, planning and execution of site preparation and earthworks shall be based on the results of a full scale or confirmatory site investigation; see DEP 34.11.00.10-Gen.

2.4 CLASSIFICATION OF SOILS AND ROCKS

Soil and rock classification should be carried out in accordance with BS 1377, ASTM D 2487, or ASTM D 2488.

2.5 SEISMIC ASPECTS

The design and eventual execution of site preparation and earthworks shall consider the relevant seismic conditions and relevant consequences prevalent in the area under consideration. Applicable seismic zones can be found in DEP 34.00.01.30-Gen.

Liquefaction potential shall be investigated.

If faults and or other geological disturbances exist at the site, their long-term behaviour, and where applicable, their behaviour during earthquake shall be investigated by a qualified expert. The conclusions of this investigation shall be taken into consideration for the subsequent earthworks, including foundation and structural design.

3. SITE PREPARATION

3.1 GENERAL

Site preparation activities may comprise one or more of the following activities:

- demolition and site clearance (3.2);
- stripping of topsoil (3.3);
- rerouting of existing ditches and water courses (3.4);
- treatment of underground cavities (3.5);
- treatment of subsoils (3.6);
- provision of site drainage (3.7);
- provision of temporary access (3.8)
- lay-down and stockpile areas (3.9);
- scour protection, coastal defence and river works (3.10);
- earthworks (4).

3.2 DEMOLITION AND SITE CLEARANCE

Demolition work may be required, depending on the nature of the existing area and planned usage. Typical demolition activities may include removal of buildings, structures, culverts, roads, pavements, aboveground and underground services and associated foundations.

Careful consideration shall be given to the treatment of existing piles to ensure the proposed tank foundation or in general any proposed foundation shall perform adequately and that there is no risk of contamination. The remnants of existing piles may result in unacceptable differential settlements if not accommodated in the design. Removal of some or all of existing piles may result in the creation of preferential seepage paths allowing seepage of contaminants into the backfill of the removed pile shaft and into the surrounding ground and groundwater. Guidance on the design of tank foundations is provided in (5.5.3).

Care shall also be taken to ensure the site clearance and demolition work does not adversely affect or diminish the capacity of the foundation of the existing adjacent buildings/structures.

Site clearance shall include the taking down or grubbing up and immediate removal from site, of surface obstructions and unsuitable materials such as debris, rubble, gates, fences, shrubs, hedges, trees, stumps and roots.

The Principal may elect to leave in place certain items such as fences, shrubs, trees, or other features required for incorporation in landscaping. The Contractor shall, under such circumstances, be responsible for the care and protection of such items during the course of the works.

Demolition and site clearance works shall be planned and executed in a manner such that the safety of personnel, the work and adjacent property is guaranteed and a minimum of inconvenience is caused.

All necessary temporary works such as safety barriers, fences, safety nets, coverings and screens shall be provided to protect personnel and adjacent property from injury or damage caused by demolition operations.

All necessary dead or raking shores to support existing work shall be provided during demolition and associated operations.

All materials arising from demolition and site clearance work, except materials specified for re-use or as otherwise specified by the Principal, shall be removed from the proposed tank farm site or any site as appropriate as the work progresses and shall be disposed of in an agreed and responsible manner. Materials removed from the facility to an external

repository shall be disposed of in accordance with all applicable local and national laws and regulations. Such materials should not be permitted to accumulate and the site shall be maintained in a safe and workmanlike condition at all times.

Materials required for re-use shall be stored in a safe location and protected from damage or deterioration as agreed with the Principal.

Segregation and quantification of hazardous and non-hazardous waste material, and certified disposal of hazardous material shall be carried out in accordance with applicable legislation.

3.3 STRIPPING OF TOPSOIL

All areas to be excavated, filled or constructed upon should be stripped entirely of topsoil to a suitable depth below ground level, as indicated by the site investigation and agreed with the Principal.

Turf intended for re-use, as agreed with the Principal, shall be taken up in strips of a constant breadth with an approved turfing tool. Turf intended for re-use shall be removed to a location approved by the Principal where it shall be neatly stacked and regularly watered and tended until required for relaying.

Topsoil intended for re-use, as agreed with the Principal, shall be stock-piled and/or spread at approved locations in limited heights as appropriate to the region to avoid detrimental affects to its chemical and biological composition.

The remaining topsoil and turf shall be disposed of at locations agreed by the Principal, in accordance with national and local laws and regulations.

3.4 REROUTING OF EXISTING DITCHES AND WATER COURSES

Existing drainage ditches and water courses shall be temporarily or permanently rerouted, where necessary, in order to prevent:

- unacceptable rise or fall in ground water level in, and flooding of, the site;
- unacceptable changes to the hydrological and hydrogeological regimes of the adjacent areas.

Rerouting shall be carried out in accordance with the requirements of DEP 34.13.20.31-Gen. and DEP 34.14.20.31-Gen.

3.5 TREATMENT OF UNDERGROUND CAVITIES

If naturally occurring cavities (caverns, underground watercourses, etc.) or man-made cavities (mine workings, wells, cesspools, basements, etc.) are detected under or adjacent to the construction area, they shall be suitably investigated and treated in consultation with the Principal.

Further guidance regarding ground improvement techniques may be found in BS 6031.

3.6 TREATMENT OF SUBSOILS

If the existing engineering properties of the ground may result in liquefaction, unacceptable total and/or differential settlements or bearing capacity failure, consideration shall be paid to the use of ground improvement methods rather than deep foundations.

Further guidance regarding ground improvement techniques may be found in BS 6031.

3.7 PROVISION OF SITE DRAINAGE

Adequate drainage systems shall be provided to accommodate ground water and surface water run off.

Permanent surface water runoff drainage systems shall be designed in accordance with the requirements of DEP 34.13.20.31-Gen. and DEP 34.14.20.31-Gen.

Further guidance regarding items including site drainage, including temporary water run-off drains and subsoil water drains, may be found in BS 6031.

3.8 PROVISION OF TEMPORARY ACCESS

Suitable temporary roads shall be laid to provide all-weather access for all construction activities. The temporary roads should preferably coincide with the planned permanent routes.

Temporary access shall be designed to suit climatic, traffic and subsoil conditions. The effects to be taken into consideration shall include heavy traffic along temporary roads, e.g., propagation of dust clouds, damage to existing structures by vibrations, disturbance of surface and subsurface drainage. Where necessary, suitable precautions shall be taken.

Traffic routing schemes should be established, especially for large-scale works.

3.9 LAY-DOWN AND STOCKPILE AREAS

Permanent lay-down areas normally located inside battery or module limits for the laying down of bundles, trays, columns, flares and other structures during shutdown or maintenance of plants shall be designed and constructed in accordance with DEP 34.13.20.31-Gen.

All temporary lay-down areas required for construction activities should be suitably levelled and provided with suitable access, temporary paving and surface drainage as necessary. These temporary lay-down areas shall be located within the designated Contractor's area. These areas shall be drained so that stored materials remain dry at all times, and surface as well as subsurface drainage may be required (3.7). Proper access to these areas for lifting equipment and construction vehicles shall be maintained.

The method of stockpiling of materials shall take into consideration all requirements for protection from the weather and from contamination through mixing. Stockpiles shall be maintained in a stable condition and work operations shall ensure safety to operators. Further guidance regarding stockpiling materials may be found in BS 6031.

3.10 SCOUR PROTECTION, COASTAL DEFENCE AND RIVER WORKS

When a facility location is selected, consideration shall be paid to protection of the proposed site from detrimental external hydraulic processes. Potential external water related hazards to a location include flooding from hinterland run-off, storm damage from the sea, flooding from high water levels and waves from the sea and/or rivers as well as erosion of riverbanks or the coastline from rivers or the sea. Protection measures to be considered are hinterland drainage channels, coastal sea defence, and coastline and/or riverbank heightening and stabilization.

Measures shall be taken to ensure the required protection with the least disturbance to the environment and to current and future land use. Typically, an Environmental Impact Assessment should be carried out for a proposed new facility and should identify sensitive environmental areas, give recommendations on minimising the impact from construction, and identify any specific social impacts or legal requirements. Where feasible, these recommendations should be adopted.

When analysing the problem, hydraulic, morphological, geological, geotechnical coastal engineering and environmental aspects shall be considered.

The hydraulic aspects shall include waves, currents, tides, seiches, storm surges (due to changes in barometric pressure and strong winds), tsunamis, rainfall intensities, storm water run-off, river discharges, floating debris, icing, the basic bathymetry and topography of the area as well as other features and processes specific to a location.

Typical morphological aspects that shall be studied include sediment transport (rate and direction), erosion and sedimentation patterns, changes in shoreline alignment (seasonal fluctuations as well as long-term development) and changes in meander pattern and bed slope of rivers.

Rises in sea level resulting from climatic changes as well as subsidence of the land through tectonic movements (if any) and other processes shall be assessed.

The geotechnical aspects shall include the slope stability of embankments, the bearing capacity and stability of the subsoil and the likelihood of settlements and erosion.

The selection of design criteria for the design of hydraulic related earthworks and structures shall be based on an assessment of anticipated events, the risks to the facility and the total lifetime costs. The certainty with which events can be predicted will be dictated by the quantity and quality of available data. Design criteria shall be based on an analysis of this data and take into consideration combinations of hydraulic parameters that can reasonably be assumed to occur simultaneously. The critical combination that results at the structure site shall be adopted for design purposes.

For the design of coastal structures, the following references may be used:

- BS 6349 - Code of practice for marine structures;
- CUR Report 154 - "Manual on the use of rock in coastal and shoreline engineering";
- "Coastal Engineering Manual", of the US Army Coastal Engineering Research Center.

Ways of minimising the life cycle cost shall be considered, including an assessment of the availability and cost of construction materials, construction methods, availability and cost of the Contractor's equipment, and the lifetime maintenance costs. The assessment of the lifetime maintenance cost shall consider the frequency of occurrence of adopted design events with other reasonable combinations and the resulting damage potential to the proposed facility. The impact on the temporary and long-term functionality of the facility and the reparability of the hydraulic structures and damaged parts of the facility resulting from the considered combinations should be assessed in order to determine the optimal economic, technical and safe solution. If an economic evaluation is not carried out, the critical combination having a return period of 100 years may be adopted for design purposes, subject to the approval of the Principal.

4. EARTHWORKS

4.1 GENERAL

Earthworks are required to establish the general shape of the site on which the facility is to be constructed. They comprise the formation of on-plot and off-plot areas, road alignments, flood protection bunds and drainage channels. Where suitable material is not available on site, material shall be imported. Unsuitable material arising from site works shall either be stored at a designated location on site, or shall be disposed of in a responsible manner in accordance with all applicable local and national laws and regulations. In some area, the final grade or level may not be formed as finishing work may be scheduled at a later date.

To minimise the import and export of material to and from a site, materials identified in the site investigation shall be considered for re-use. The approximate quantities of materials shall be determined and laboratory testing shall be carried out to determine the engineering properties and mineralogical composition of potential sources of fill. Site construction trials shall be carried out to satisfactorily demonstrate that the proposed equipment, compaction methods and compliance testing programme fulfil the design requirements.

BS 6031 extensively describes the design of cuttings, excavations, embankments and general fill, execution of excavation, filling and compaction. Also, the design and construction of trenches and pits is covered in this standard. Earthworks shall therefore be designed, planned and executed in accordance with BS 6031. However, suggested procedures and indicative compaction methods are provided in (Appendix 1) and (Appendix 2).

Protection of all earthworks and the control of storm water run-off and groundwater shall be included in the design as specified in DEP 34.13.20.31-Gen. Appropriate temporary and permanent drainage systems shall be installed and any erosion of earthworks shall be controlled. The design shall ensure the avoidance of separation of fill materials (e.g. migration of fines), and geotextiles may be considered for this purpose.

At locations where uncommon materials are considered for use, e.g., crushed rockfill, specialist advice shall be sought and construction trials may be required to ensure an adequate design and performance of the fill.

4.2 CONTROL OF EARTHWORKS

The earthworks design shall address the functional requirements, which may include the permitted allowable total and differential settlements, minimum or maximum permeability, minimum factor of safety for stability, special measures to control erosion, etc. The specification shall define the physical characteristics of the suitable materials (grading, angle of friction, etc.) and describe the work on the basis of either an end product or a method specification.

An end product specification defines the properties of the completed earthworks, but does not define the process to achieve the final product. A method specification is prescriptive, stating amongst others the moisture content of the material as placed, the maximum thickness of each placed layer, the type of compaction equipment which shall be used, as well as the number of passes of the equipment, etc.

The specification shall include a defined compliance testing programme stating the type of testing required and the frequency of each test. It shall also provide the ranges of acceptability. Compliance testing typically comprises the sampling and testing of stockpiled material for use as fill and testing of the placed and compacted material. The properties to be tested of the stockpiled material shall be appropriate for the type of fill material and may include grading, plasticity, shear strength, angle of friction and moisture content. Testing of the placed material shall include the moisture content, bulk density and dry density, and possibly other properties.

Prior to commencement of earthworks, a method statement shall be produced and submitted to the Principal for comment. The method statement shall describe the proposed equipment, and a compaction trial covering the proposed methods of excavation (for

cuttings), placement and compaction (for embankments). The compliance testing programme shall be clearly defined in an inspection and test plan, identifying methods and equipment that shall be used to determine the specified physical and engineering properties, and provide sample Quality Control Record (QCR) sheets stating the information which shall be recorded for each compliance test. If the nuclear density probe test (NDP) is proposed (ASTM D 5195, BS 1377), a trial shall be conducted to demonstrate that the test is appropriate for the materials and to calibrate the NDP test results against conventional testing methods (e.g. sand replacement, or core cutter methods (BS 1377)). The acceptable ranges for each test result shall be identified in the inspection and test plan.

The compaction trial shall not be incorporated into the final earthworks. The results of the compaction trial shall be reported to the Principal prior to commencement of the earthworks. If any part of the compaction trial fails to comply with the requirements of the specification, the method statement shall be revised and additional trials conducted until satisfactory results are achieved.

During execution of the earthworks, compliance testing shall be carried out as specified in the inspection and test plan, and the results shall be reviewed against the specified allowable ranges. If results are noncompliant, the Contractor shall propose corrective actions to the Principal that are appropriate for the specific non-compliance.

The compliance test results shall be stored either electronically or in paper records as appropriate to the work scope and location. A summary report shall be produced identifying the areas on a plan in which controlled engineered fill has been placed. For each area, the summary report shall reference the exact specification to which the work was carried out and present a summary of the compliance test results.

4.3 PREPARATION OF SOIL BEARING FOUNDATIONS

If calculations show that ground conditions may produce unacceptable settlements, many well established methods are available to improve the situation, some of which are described in (3). In principle, consideration shall be given to the simplest and most cost effective methods of improving the ground conditions. However, where time constraints preclude the least expensive solution, more sophisticated and expensive methods may be required. The design of ground improvement works is a specialist area that requires a thorough understanding of the ground conditions and capabilities of appropriate ground improvement techniques to ensure the appropriate solution is adopted. In some instances, ground conditions and/or time constraints will result in a piled foundation.

Preloading of compressible areas through the construction of temporary embankments is a simple and cost effective method of achieving consolidation. Further economies can be achieved if the material used to form the preload embankment comprises suitable materials that can be re-used in construction upon removal. During construction of the preload embankment and the subsequent settlement period, a monitoring system shall be installed. At least three settlement beacons shall be installed along the future axis of each foundation. Survey data shall be recorded on a regular basis in accordance with a predefined programme to determine settlement. In addition, for more sensitive soils, or at proposed locations of critical structures, other parameters may require to be measured which may include pore water pressures, settlements at depth and surface settlement profiles.

The recorded data shall be analysed and reference shall be made to the predicted consolidation rate. If results indicate the consolidation rate is slower, or the settlements are larger than predicted, measures shall be taken to gain a better understanding of the ground conditions and settlement characteristics so that modifications can be implemented if necessary. The foundation design can be verified by monitoring settlements and other specified parameters during the construction and preload period.

The duration of the preload period depends on the subsoil condition that will dictate the duration of the hydrodynamic period. In some cases, where the calculated preload period is too long, methods of accelerating the settlement by installing of vertical drains may be considered. However, this is only permitted at locations where there is no risk of

groundwater and soil contaminations, with the vertical drains providing a preferential drainage path.

Other alternative solutions to treat poor ground include the use of special fills, e.g., lime or cement stabilisation, ground replacement (excavation of poor material and replacement with compacted suitable material, see (Appendix 11)), soil mixing, dynamic compaction, vibro compaction, vibro replacement and pressure grouting. In all cases, the main principles shall apply:

- design of ground improvement system;
- production of detailed method statement;
- controlled work including construction trials with quality records produced;
- monitoring and post treatment compliance testing to verify design.

4.4 MOUNDS FOR BURIED VESSELS

Mounds for buried liquid or gas-containing vessels shall be constructed in accordance with DEP 34.51.11.30-Gen.

5. BUNDED AREAS AND FOUNDATIONS FOR TANKS

5.1 BUNDED AREAS FOR TANKS

5.1.1 General

Amended per
Circular 60/08

The material and type of the banded area shall be assessed in relation to the function of the tank, with due account being taken of geotechnical aspects. The main factors to take into consideration are the potential environmental and economic risks that may result from a tank spillage. If water or non-very-toxic fluids are stored, and there is no potential risk to surrounding areas in the event of a spillage, the requirements for the bund may be less demanding than if hydrocarbons or very toxic chemicals are stored, or where process equipment or other sensitive assets may be placed at risk.

A finishing layer that shall be appropriate for the location shall cover the banded area. If contaminants are being stored, impermeable membranes shall be required to prevent egress from the banded area in the event of a tank leak. In all cases, erosion control on both sides of the bund shall be accommodated in the design to prevent deterioration in the bund resulting from anticipated heavy rainfalls, or during a leakage of the tank.

5.1.2 Functions and requirements

The function of a banded area for tanks and its associated drainage system is to contain potential spillages from any tank in the banded area so that subsequent damage to adjacent tanks, their contents and the surrounding areas is minimized.

Tanks containing water, e.g., fire-, boiler feed-, condensate-, blow down-, raw water, etc., do not require a banded area unless they are located adjacent to areas where the economic or safety impact resulting from a spillage would be unacceptable.

Layout sketches for various tank farm configurations are shown in (Appendix 7). The European Model Code of Safe Practice, Part II may also be used for determining layouts and safety distances for marketing and distribution facilities.

The net capacity of a tank bund (banded area) formed within the main fire wall should be sufficient to provide reasonable protection against possible escape of products beyond those walls. Reasonable protection in terms of volume is considered to be when the bund has a net volume capacity equal to the volume of the largest tank in the banded area. The total volume of the bund should be calculated by adding to this net volume capacity the following volumes insofar as these volumes are below the top level of the surrounding bund wall:

- the volume of the tanks except that of the largest tank in that bund;
- the volume of all tank pads;
- the volume of all intermediate fire walls and the low level piping.

In larger banded areas, intermediate firewalls may be designed (normally not higher than 0.6 m) to act as firebreaks and to divide the tankage into groups of a convenient size.

Guidance for the design of tank farms is provided in API STD 620. However, bund walls shall be designed in accordance with BS 8007 unless otherwise indicated by the Principal or otherwise stated in this DEP.

Drainage systems shall be designed and constructed in accordance with DEP 34.14.20.31-Gen.

5.2 IMPOUNDING BASIN

If land availability and space restrictions do not permit the banded area design to satisfy the requirements of (5.1), a suitable depression or a purpose-made impounding basin may be used conveniently situated elsewhere within the installation. This impounding basin shall be

connected to the tank farm(s) by an adequately sized underground drain line, with open catch basins at both ends.

The need to incorporate pumping facilities or discharge valves to empty the contents of the impounding basin shall be assessed.

5.3 BUNDS AND BUNDWALLS

5.3.1 Design of bunds

Bunds should be designed and constructed as a liquid-tight or essentially impermeable earthen or clay structure with a slope not steeper than 1:1. For details, see Standard Drawing S 12.002. In some instances, reinforced concrete or steel retaining walls may be more economic or practical especially where embedded cut-off walls are required to prevent seepage. Examples of typical retaining walls are presented in (Appendix 8).

For operational and maintenance access, at least two stairways (preferably made of concrete) shall be provided for each bunded area. These stairways should be located at opposite sides of the bunded area.

The crest elevation of the earthen bund wall shall have a minimum width of 0.60 m. The height of the bund wall above the bund floor shall be determined from the summation of the following heights:

- a) required net capacity of the bund;
- b) maximum determined bund settlements with respect to the tank foundation during the design life;
- c) a freeboard of 0.30 m.

The freeboard is included to allow for the following affects that may or may not occur:

- a) storm surge (0.30 m);
- b) foam containment;
- c) collection of rainwater;
- d) standing water in bunded area (0.15 m);
- e) sloshing resulting from sudden collapse of a storage tank.

The stability of the bund and its foundation (i.e. its horizontal stability against sliding, as well as the stability of slopes and subsoil) shall be analysed by taking as a design basis the bunded area being completely filled with water (i.e. to bund crest). The analysis shall assume long-term design parameters (i.e. effective shear strength parameters ϕ' , c'). The influence of possible liquid flow through and/or under the bund wall on the stability shall be investigated. Reference is made to DEP 34.11.00.12-Gen. for the stability analysis and the soil deformation aspects of bund wall design.

Design safety factors shall not be less than 1.3.

The penetration of earthen bund walls and the underlying soil with service line (i.e. pipes, cables, etc.) should be avoided. If penetrations are unavoidable, protective pipe sleeves with sealed, flexible collars at the upstream end shall be used. Free space shall be provided between the protective pipe sleeve and the service line to allow for the expected settlement. The sealing material between the pipe and sleeve shall be malleable to remain functional in the event of movements.

Filter cloth shall be placed around rigid elements where there is a danger of bund wall materials being washed out; see Standard Drawing S 12.002. If more than one service line penetrates the earthen bund walls, structural elements forming an integral part of the bund wall should be used.

The proposed design of these subsurface or penetrating service lines and the protection measures is subject to the approval of the Principal.

Bunds constructed from materials susceptible to erosion, and covered by a sand, bitumen and cement mix or a clay cover, shall be provided with toe drainage and a toe filter in order to prevent damage to the cover by build-up of pore water pressure within the bund wall; see Standard Drawing S 12.002. Care shall be taken to ensure that this toe drainage does not create a seepage path in the event of a spill.

5.3.2 Bund materials

To ensure the liquid tightness of the bund, it shall be constructed partly or wholly of impermeable or low-permeable materials. A bund constructed entirely with clay is normally preferred. In locations where clay is scarce or not available, the following may be considered:

- a sand body with a clay cover of at least 300 mm thick;
- a sand body covered by a durable mix of sand, bitumen and cement, at least 75 mm thick.

5.3.3 Construction of bunds

Site preparation for bunds shall be carried out in accordance with (3).

Placing and compaction of materials shall be carried out in accordance with (4).

The surface protection measures required for earth bunds are described in (5.3.4).

5.3.4 Bund covering (erosion protection)

If the climate permits, solid clay or clay-covered bund walls should be covered with a layer of topsoil 150 mm thick and grassed. Topsoil shall be lightly compacted and raked prior to the sowing of grass. If necessary, turf shall be laid and pegged in place. The grass type should be a locally available variety and shall provide strong dense roots and growth. It shall require minimal maintenance. The use of fertilisers and mulching, as well as requirements for watering and cutting of grass, shall be agreed with the Principal. Grass that grows tall and dries out, forming a fire hazard, is not acceptable.

The covering comprising a mix of sand, bitumen and cement shall be well compacted and given a smooth finish. The durability of these covers shall be proven through a trial mix and application. Cracks shall be repaired by removing the cover back to sound material and filling the cracks with molten bitumen. Larger rejected areas shall be trimmed back to original base and repaired with a minimum of a 75 mm thick layer of the sand, bitumen and cement mix.

Rigid coverings (e.g., concrete) shall not be used for earthen bunds because of the increased risk of uncontrolled cracking followed by washout of bund material from under the cover and subsequent loss of integrity.

For bunds designed to contain chemical or light products, linings (Low Density Polyethylene (LDPE) or High Density Polyethylene (HDPE)) shall be introduced both on the surface of the bunded area and in the bund itself.

5.3.5 Structural elements

Structural elements such as reinforced concrete retaining walls or steel sheet piles may be used where many pipes cross the bund or where there are space restrictions. However, if this type of bund is adopted, special attention shall be paid to the following:

- stability of the wall;
- panel joints (water stops) e.g. by means of hydrocarbon resistant sealant or a flexible polyurethane joint filler;
- pipe penetrations;
- subsurface drainage of the tank bund;
- flow and void formation under the structural retaining elements;

- consequences of settlements and the effect of the relative rigidity of the structures and bunds;
- deflection or deformation of the structure during the containment of liquids;
- means of access for operational and maintenance purposes.

A waling may have to be included in sheet piled bund walls in order to reduce uneven deformation or deflections of the wall when retaining liquid that could lead to increased seepage through the joints.

5.4 SURFACE FINISHES OF BUNDED AREAS

Surface finishes in banded areas shall have a minimum gradient of 1:100 towards the perimeter drainage. Drainage systems shall be designed and constructed in accordance with DEP 34.14.20.31-Gen.

The permeability of the surface finishes of banded areas depends on the product stored in the tanks, on local circumstances and on legal requirements that may in some instances require a completely impermeable surface. The proposed design for the surface finish is subject to the approval of the Principal.

In the design of surface finishes of banded areas, the following requirements shall be adhered to:

- all parts of the surface of the banded area shall meet the minimum permeability requirements mentioned above;
- the surface finish and drainage system shall be durable;
- the surface shall access by and support of heavy equipment for operation and maintenance activities.

The surface finish shall not be continued underneath the tank pads. This is to avoid the possibility of the tank pad losing inherent strength by becoming saturated from a tank leakage or penetrating rainwater. The surface finish shall be connected to the seal of the tank base.

A layer of 75 mm pea gravel, concrete tiles or equivalent should protect membrane liners.

Tall dry grass or other vegetation covering the bund floor is not acceptable because of the associated fire hazard.

5.5 TANK FOUNDATIONS

5.5.1 General

The type of foundation for a vertical storage tank shall be selected in relation to the function of the tank, with due account being taken of geotechnical aspects. This assessment shall ensure the tank remains stable and that the total and differential settlements remain within serviceable limits and connecting piping is not damaged.

Foundations for vertical storage tanks should take the form of a tank pad constructed with durable, inert, granular materials, such as crushed rock, coarse sand, etc.

NOTE: Sands from volcanic origin and carbonate sands are not suitable.

The tank foundation shall be covered with a surface finishing layer (5.4).

Atmospheric storage tanks are rarely founded on piled-concrete foundations, unless the predicted differential and total settlements and tilt exceed the limits set for the tank in question; see (6). In exceptional cases for small tanks, a concrete raft or ring beam is sometimes used as a foundation.

Anchor bolts (Standard Drawing S 51.046) may be required to hold the tank wall down, e.g., in earthquake, hurricane or typhoon areas, or where the internal tank pressure is high (e.g. for Butt-welded High-pressurized Cone-roof (BHC) tanks), in which case a reinforced

concrete ring beam or other appropriate solutions may be required; see (5.7) and DEP 34.51.01.31-Gen.

The following paragraphs deal only with tank foundations comprising granular materials (i.e. sand, crushed rock, etc.). The use of non-granular materials for tank foundations, or the use of concrete raft foundations that are not piled, is not allowed unless specifically approved by the Principal.

5.5.2 Tank foundation functions and requirements

The functions of a tank foundation are:

- to spread and transfer the load from the tank and its contents via the tank foundation body and shoulder to the subgrade so that the resulting ultimately expected settlements, both total and differential, remain within the allowable limits;
- to raise the tank bottom above ground water, capillary water, surface water and the standing fluid level resulting from minor spillages;
- to provide a smooth surface with sufficient bearing capacity and stability for tank construction and operation;
- to channel rain water away from tank shell and tank bottom quickly;
- to provide a raised elevation to the tank base so as to enhance the NPSH (net positive suction head) of pumps.

The requirements for the shoulder of the tank foundations are:

- to provide sufficient lateral support to the tank foundation under all conditions. The shoulder shall be capable of providing sufficient lateral support during construction, operation and maintenance activities;
- to resist edge settlement (cutting) beneath the tank shell;
- to resist wash-out of the tank foundation as a result of tank bottom leakages and possible ingress of rain water;
- to resist erosion by wind and/or water.

5.5.3 Design of tank foundations

5.5.3.1 Site investigations

The tank foundation profile (e.g. cone up, cone down, dimensions and shoulders) shall be chosen according to the stiffness and homogeneity of the underlying soils and the results of settlement and stability calculations. These calculations shall be based on the results of a soil investigation and are subject to the approval of the Principal. The properties of the existing subsoil shall be thoroughly investigated. The site investigation shall include a study to establish the history of the site and identify any natural processes or previous development that may influence the characteristics of the ground (5.7.4).

The techniques employed for the site investigation depend on the characteristics of the site to be investigated. Common methods of investigation include Cone Penetration Tests (CPTs), boreholes, and in-situ and laboratory testing. The purpose of the investigation is to determine the physical, mechanical and chemical properties of the underlying strata.

The type and number of laboratory tests shall be as specified in DEP 34.11.00.10-Gen. The investigation shall be carried out by a qualified and accredited geotechnical contractor which has experience with local soils and with tank foundations.

The report shall as a minimum include advice on the following items:

- whether it is possible to build the tank according to DEP requirements at the proposed location;
- requirements for soil improvement including the type and extent of soil improvement;
- the influence of the excavations (if required) and the load of the new tanks on adjacent tanks or structures;

- the influence of wind and earthquakes on the tank, tank contents, tank foundation and subsoil in relation to the possible liquefaction of the subsoil;
- the time-settlement behaviour of tank foundation and tank;
- the required materials for use in the tank shoulder (e.g. sand, crushed rock etc.) and the width of the various materials;
- the required compaction of tank foundation and shoulder;
- the maximum tank height;
- a proposal for a hydrostatic testing scheme including any restrictions on the rate of filling during hydrostatic testing to ensure the stability of the tank foundation. The analysis shall take into account the possible effects of increases in pore water pressure, settlement and creep. Details shall be given of a monitoring programme to be carried out during the hydrostatic test and the validation exercise to confirm that the performance of the ground is in line with predictions. The monitoring programme should also state the actions that shall be taken to ensure the stability of the tank and its foundation system, if limits are exceeded.

Note: Hydrostatic testing, generally done with water, normally establishes the maximum load for the tank and tank foundation. If the specific density of the product in the tank will be higher than that of water, the extra load of the tank shall be extrapolated from the hydrostatic test.

The required number of CPTs or borings depends on the diameter of the proposed tank and the variation of the subsoil layers. The minimum required number of CPTs or borings is:

- 5, for tanks up to 12 m diameter;
- 9, for tanks larger than 12 m and up to 40 m diameter;
- 13, for tanks larger than 40 m diameter.

The CPTs shall be carried out on the tank foundation circumference at approximately 10 m centre to centre and one near the centre, together with one or more borings. The required depth of CPTs and borings normally depends on the tank diameter but shall always include the compressible or weak layers. As a minimum, the depth of the exploratory holes shall be the greater of either the diameter of the tank or 25 m depth. In some instances, further penetration may be prevented before this minimum depth due to increased strength or density of the ground, see (Appendix 9), or due to encountering rock or an obstruction. In such instances, the reason for refusal shall be determined and approval shall be sought from the Principal to terminate the investigation at the reduced depth.

5.5.3.2 Design parameters

The tank foundation shall be designed as a flexible foundation with sufficient strength to ensure that it is capable of:

- absorbing subsoil deformations to ensure deformations in the tank base remain within serviceable limits;
- distributing the possible unequal foundation pressures.

The relative rigidity of the tank wall with respect to the bottom slab and the subsoil shall be taken into consideration. Total rigidity of the tank foundation (i.e. concrete slab resting on soil) and/or rigid elements in, around, and under the tank shall be avoided.

Under certain circumstances, soil improvement (4.3) under the tank foundation shall be considered in order to:

- provide a foundation with sufficient strength;
- reduce large differential settlements and total settlements;
- avoid or reduce earthquake induce liquefaction potential.

The minimum elevation of the tank foundation at the base of the tank shall be 0.60 m above the adjacent terrain. The predicted long-term settlement shall be added to this value.

Consideration should be paid, where possible, to the re-levelling of a tank (under the tank shell), rather than the complete jacking of a tank, to compensate for differential settlements or tilt because this may be technically acceptable and often proves to be the most economic solution.

Reference is made to DEP 34.11.00.12-Gen. and DEP 34.11.00.10-Gen. for the geotechnical aspects of settlements and to DEP 64.51.01.31-Gen. for the structural limits.

Drain pipes shall be placed around the circumference of the foundation (if completely closed) in order to detect leakages and to prevent the build up of pore pressure due to liquid accumulating in the tank pad.

Geotextile shall be installed under the surface finishing to the shoulder if there is a possibility of washout of fine materials; see Standard Drawing S 12.001. Geotextile type Nicilon No. 66339 or equivalent shall be used.

A stability analysis of the shoulder and tank foundation shall take into account the following aspects:

- wind loads;
- earthquake loads;
- blast waves (if so required by local law);
- initial height above adjacent ground level;
- highest possible ground water level;
- angle of slope of the shoulder;
- geotechnical properties of the tank foundation materials and subsoil;
- load provided by the tank and its contents during hydrostatic testing, operation and maintenance.

The liquefaction potential of the subsoil due to earthquake or other dynamic load source shall be investigated.

5.5.3.3 Tank foundation shoulders

The shoulder width shall be selected so that the stability of the foundation, shoulder and subsoil is ensured. The minimum width of the tank foundation shoulder (S) depends on several aspects:

- height of the tank (H);
- density of the product (for oil products, water filling in hydrostatic test is normative);
- the type of tank foundation (no crushed stone ring, crushed stone ring, piled etc.);
- the slope of the tank foundation edge;
- the height of the tank foundation (T).

The following limiting values for the shoulder width and thickness shall be adopted for tanks of various heights:

Tank foundations without crushed stone ring:

H	S (minimum)	T (maximum)
[m]	[m]	[m]
10	1.00	1.20
20	1.50	1.00

Tank foundations with crushed stone ring:

H	S	T
	(minimum)	(maximum)
[m]	[m]	[m]
10	1.00	1.50
20	1.50	1.00

H = height of tank;

S = width of tank shoulder measured from annular plate edge;

T = height of tank foundation before hydrostatic testing.

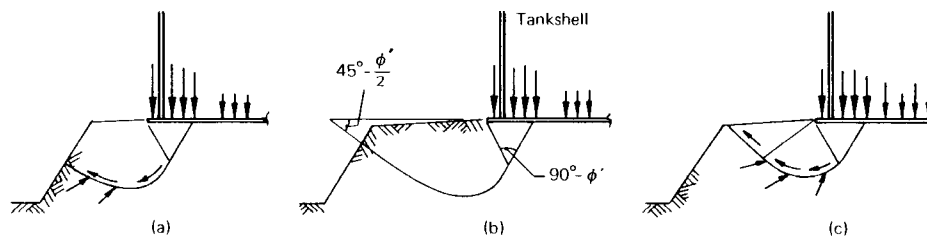
The shoulder shall have a gradient of 1:10. The slope to the shoulder shall not exceed a gradient of 1:1.5.

To properly compact a foundation comprising entirely of sand for a tank with a diameter (D), the width of the tank foundation during construction of the foundation and the tank shall be larger than the final dimensions. The minimum dimension during compaction shall be $D + 6S$. The tank foundation can be trimmed back to the correct size on completion of construction of the tank.

For a tank foundation with a crushed stone ring, the above-mentioned enlargement is not necessary.

The wide compacted shoulder provides lateral support to the concentrations of stress in the soil below the tank shell; see Figure 1.

Figure 1 Tank Foundation Shoulders



- Illustrates a shoulder that is too narrow, where soil movement under the shell or even a slip failure is likely to occur.
- Illustrates a shoulder that is wide enough, but not sufficiently compacted. Consequently, the angle of internal friction of the shoulder material is low and a slip failure could occur. Settlements and differential settlements will also be experienced.
- Illustrates a wide, sufficiently compacted shoulder with no risk of soil movement or a slip failure of the shoulder.

In general, a slope stability failure occurs when the resisting forces are lower than the forces generating the failure (i.e. the factor of safety falls below unity). To reduce the probability of failure, a number of actions can be taken. The shoulder can be widened and thereby the rupture path will be lengthened and more resistance to failure will be achieved. Alternatively, crushed stone that has a high angle of internal friction that results in a higher resistance to slip failure could be used. However, it is possible that a failure may result through the underlying soil where, although the failure path is longer, the resistance is lower due to the lower angle of internal friction of the soils. Care shall be taken in the analysis to ensure that the potential rupture path with the lowest factor of safety is identified and that this meets the minimum design requirements.

Foundation shoulders should be wide enough and properly compacted. In particular, a high density shall be achieved through compaction for the material under the annular plate and under the tank shell, as well as a short distance outside the tank footprint to ensure the high concentrations of load are evenly distributed at a shallow angle through the shoulder and into the underlying ground. This will result in spreading the load from the tank shell and minimising the bearing pressure on the subsoil. This is especially important on soft silts or clays.

5.5.4 Materials

5.5.4.1 Tank foundation body

The tank foundation body shall be constructed from clean granular materials which meet the following requirements:

- not crushable;
- low compressibility ($C^4_{10} > 500$);
- high friction properties ($\phi' > 35^\circ$, $c' = 0$);
- low silt content ($\leq 10\%$ m/m);
- free draining ($k \geq 1 \times 10^{-4}$ m/s);
- insensitive to weathering, oxidation, chemical changes, changes in mechanical properties;
- easy to compact
- not sensitive to liquefaction (especially when constructed in earthquake zones).

Well-compacted sand, especially coarse sand, amply meets these requirements providing the chemical and mechanical stability of the minerals is guaranteed.

In order to prevent capillary rise of the groundwater the upper 200 mm of the tank foundation shall comprise coarse sand.

The specifications for sands to be used in tank foundation construction, as well as the specifications for sands for soil improvement, are given in (Appendix 13).

5.5.4.2 Gravel backfill under tank shell and annular plate

A 'foundation ring' for the tank foundation shoulder and a zone under the tank shell and tank bottom constructed from hard crushed rock is preferable to sand. This material is stronger and has a greater resistance to edge settlement and damage during construction and erosion. The specification for crushed stone to be used in tank foundation construction is given in (Appendix 13).

The tank foundation body inside this ring may comprise finer granular materials such as sand. However, a filter cloth (geotextile) shall be installed between the sand body and shoulder material in order to prevent washout; see Detail 1 on Standard Drawing S 12.001.

Granular material as specified in (5.5.4.1) for the tank foundation body may also be used for the "foundation ring" provided special attention is given to compaction and protection from damage during tank construction. In addition, measures shall be taken to ensure the material under the tank shell is not disturbed, as this could result in edge settlement and possible tank deformation or failure. Therefore, the tank foundation shall be built larger than the original dimensions; see (5.5.3.3).

The use of a concrete ring, in order to prevent edge cutting or to accommodate anchor bolts, requires special attention because poor construction may damage the tank bottom and influence the settlement behaviour of the shell.

5.5.5 Construction

Site preparation shall be carried out in accordance with (3).

Ground improvement as well as the placement and compaction of foundation materials shall be in accordance with (4). Typical processes for construction of tank foundations are presented in (Appendix 12). During construction of the tank foundation, compliance testing shall be performed on each 0.3 m thick layer of placed and compacted material by means of one of the following test methods:

- the Nuclear Density Gauge method; see (4.2);
- the core-cutter method;
- sand replacement method.

The number of tests to be taken **per layer** shall be the same as given for the number of CPTs to be taken on completion of the tank foundation outlined in (Appendix 14).

When the construction of the tank foundation is completed, verification testing shall be carried out to confirm the total tank foundation meets the requirements of the design. The CPT shall be used for this verification exercise and shall as a minimum extend through the tank foundation material and any underlying strata which may have been subject to ground improvement. The frequency of testing required is presented in (Appendix 14). Typical minimum cone resistances for soil compaction are presented in (Appendix 15). If the only locally available and economic test apparatus to perform this verification exercise is a DCPT (Dutch Cone Penetration Test), then the requirements of (Appendix 5) shall be adhered to.

The penetration depth for the verification testing shall extend, as a minimum, to the larger of the following values:

- 5 m below the surface of the tank foundation;
- the full extent of the substrata which has been subject to ground improvement.

This is to ensure the engineering properties of the subsoil beneath the tank foundation meet the minimum values adopted in the design. Should the verification test fail to meet the required investigation depth, the reason for refusal shall be determined and proposals for corrective action shall be submitted to the Principal for approval.

Should the recorded results of the verifications tests fail to meet the minimum design values adopted for the tank foundation and the substrata, the reason for the reduced values shall be determined and proposals for corrective action shall be submitted to the Principal for approval.

5.5.6 Tank foundation finishes

The purpose of the surface finishing layer under the tank foundation is:

- to act as a barrier to corrosion promoted by water or water vapour, together with chemicals which may be present in the tank foundation or subsoil;
- to promote a uniform distribution of stress from the tank bottom to the tank foundation;
- to protect the tank foundation during construction;
- to allow thermal expansion of the tank bottom.

The function of the surface finishes of the tank foundation shoulder is to protect the foundation from damage caused by weathering, erosion and construction, operation and maintenance activities. The tank foundation and tank foundation shoulder shall be in an optimum condition during hydrostatic testing.

The surface finish layer under the tank shall be a mixture of sand and bitumen, or oiled sand for hot ($> 70^{\circ}\text{C}$) tanks. The sand of this mixture shall be clean and have a low silt content. The exact composition of the dry sand bitumen mix shall be based on the assessment of field trial results. Guidelines for the design and placement of dry sand mixes are given in (Appendix 3). Where a sand and bitumen mixture is used, the layer shall protrude slightly (approximately 150 mm) beyond the bottom plates of the tank and, after completion of the tank construction and testing, this ring shall be filled carefully with bitumen to minimize corrosion of the bottom plates.

The final levels of the placed and compacted surface finishes under the tank edge between any two points 10 m apart around the periphery of the tank shall not be greater than ± 6 mm and the envelope of peripheral surface levels shall lie within 12 mm above to 12 mm below the design levels; see BS 2654.

The tank foundation shoulder may, like bund walls, be covered by a mixture of sand, bitumen and cement or lime (i.e. a wet sand mix), a (hot) micro-asphalt mix or, if a light hydrocarbon or chemical spill is likely, a concrete covering comprising loose elements or tiles directly on a geotextile.

The surface finish to the shoulder shall have a gradient slope of 1 vertical in 10 horizontal from the underside of the annular plate of the tank, in accordance with Standard Drawing S 12.001 to avoid ingress of water under the tank after some tank settlement has occurred. The exact composition, which depends on the sand grain size distribution, particle shape, etc., shall be based on the assessment of field trial results. Guidelines for the design and placing of wet sand mixtures are presented in (Appendix 4).

5.5.7 Tank (bottom) settlement

Tank settlement shall be calculated with the tank full of water during the hydrostatic test period. The settlement during lifetime of the tank shall be calculated with full product load. As the tank is not (100 %) fully filled during its lifetime, the calculated settlements are assumed to be conservative. However, the effect of cyclic loading and unloading resulting from operations during the lifetime of the tank shall be investigated to verify this assumption.

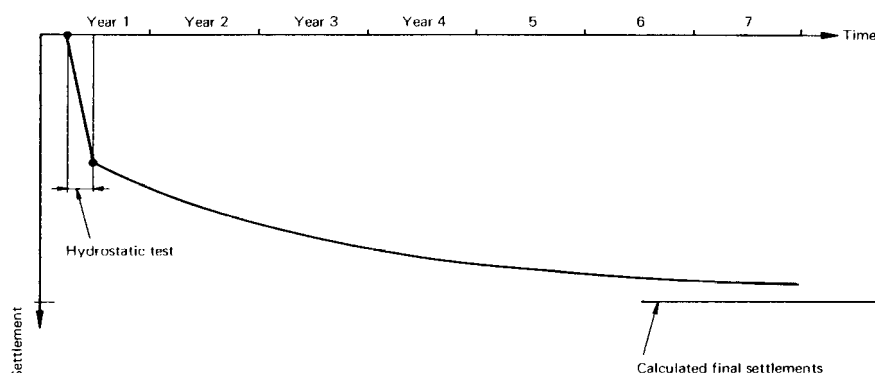
In calculating the stability of the tank and tank foundation, the tank shall be assumed to be full of a fluid whose density is the higher of either water or product.

In general, the tank centre will settle substantially more than the tank edge because of variation in stress distribution.

Typically, 30 % to 60 % of the total settlement will occur during hydrostatic testing of the tank and the remaining settlements will occur mainly during the first few years when the tank is in operation. However, this will depend on the properties of the underlying strata, variations in the elevation of groundwater, and possible freeze/thaw effects in cold climates.

It is typical for settlements to gradually become less, as can be seen in Figure 2 below:

Figure 2 Schematic Settlement Curve of Tank Edge



If no large settlements are expected, the tank shall be built with the base having a cone-down profile and a centre drain shall be provided. If large settlements are expected, a tank bottom with a cone up profile may be considered. Allowance shall be made for either fast or slow consolidation rates, whichever is the least favourable. The gradient of the bottom shall not exceed 1:120 and the maximum cone up in the centre shall not exceed 300 mm in order to reduce the possibility of rippling of the tank bottom.

If a cone up profile is adopted for the base of the tank, a side drain may be required. The decision to select either side or centre drains will be governed by the results of consolidation settlement calculations at the completion of hydro testing. If it is calculated that the cone up profile will remain after hydro testing, side drains shall be installed in addition to the centre drain. For details of side drains, see Standard Drawing S 51.088.

5.5.8 Hydrostatic testing of tanks

The hydrostatic testing procedure of tanks, which is governed to a great extent by the behaviour of the subsoil, is subject to the approval of the Principal. The hydrostatic test programme shall incorporate the advice provided in the geotechnical report described in (5.5.3.1).

The objectives of hydrostatic testing are:

- controlling consolidation of the subsoil for new tank pads;
- checking uniform settlement of the tank for renovated tank pads;
- checking stress relief in bottom-wall connection and/or vertical wall sections after repair;
- checking the out of roundness of the roof/screen versus vertical movement by floating roof tanks and floating screen tanks;
- checking leak tightness of the tank.

Depending on the objectives, one of the following options will be applicable:

- complete hydrostatic test without limitation of filling rate and intermediate hold points;
- complete hydrostatic test without limitation of filling rate;
- partial hydrostatic test;
- the filling rate and the intermediate hold points shall be determined in consultation with the civil engineering department.

Guidelines and examples of hydrostatic testing procedures for different soil types are presented in (Appendix 6).

5.6 TANK FOUNDATION WITH LEAK AND DETECTION OF LEAK MANAGEMENT SYSTEM

5.6.1 General

Bottom leakages may occur due to corrosion of either the topside or underside of the tank base, welding defects or cracks in base plates, or due to internal mechanical damage. The extent of corrosion can be detected by means of internal inspection techniques but these techniques are limited. Determining the nature (e.g. general, pitting, etc.) and extent of the bottom under side corrosion is more difficult. Therefore, in some instances where local legislation or environmental or safety issues prevail, it may be necessary to install a leak control system.

New tank pads shall be provided with a leak detection system such as a system to monitor the groundwater quality at specific intervals.

The leak control system shall not be used where products will solidify due to cooling after passing through a leak, i.e. liquid sulphur, bitumen, wax etc. For details of leak control systems, see Standard Drawing S 12.003.

Amended per
Circular 60/08

An HDPE liner shall be used in storage tanks containing hydrocarbons and/or very toxic substances that can contaminate the tank foundation and its subsoil and groundwater. HDPE may not be resistant against all chemicals, particularly acids. The durability of HDPE shall be assessed for the product to be stored in the bunded area.

5.6.2 Installation and testing

The HDPE liner (or equivalent) as shown on Standard Drawing S 12.003 comprises lap welded sheets with the overlap placed in the down slope direction towards the centre drain sump. The placement of HDPE sheets and the welding and testing of joints, etc., shall be carried out in line with the Manufacturer's instructions and procedures. In locations where there are large differences in temperature between day and night, the HDPE lining should be installed in the early morning and covered with sand to avoid excessive rippling of the sheet due to increase in temperature during the day.

The HDPE drain sump in the centre (either purpose made or standard) shall be connected via a drainpipe to an inspection sump. This inspection sump shall be placed a minimum of 0.5 m outside the toe of the tank foundation.

The primary function of the liner is to direct any product leakage that occurs through the tank bottom to the drain sump. From the drain sump, the leaked product will flow along the drainpipe to the inspection sump. To allow this flow of leaked product, the levels and gradients of each component of the leakage detection system shall comply with Standard Drawing S 12.003, Table 1. The design installation levels for each component of the leak detection system shall take into account the maximum allowable settlement of the tank bottom and tank edge, as well as the maximum allowable tilt. The design shall ensure that, given these allowable maximum movements, the leaked product shall flow through the entire system and into the inspection sump under gravity.

A 25 mm diameter HDPE test pipe shall be installed and a test shall be carried out annually. The test shall comprise the introduction of a water dye tracing fluid into the test pipe and monitoring of the inspection sump to confirm that drainage flow from the centre drain sump to the outside inspection sump is maintained.

Before the HDPE liner activities commence, the contractor shall submit an inspection procedure indicating inspections and test reports to be conducted during preparation and application of the HDPE liner for approval by the Principal. The execution plan shall include a welding test and a spark test. Welding of the HDPE liner is only permitted if the following conditions are met:

- temperature of HDPE liner and ambient temperature are higher than 5 °C;
- wind speed is lower than 8 m/s (5 Beaufort);
- relative humidity is lower than 90 %;
- no water is present on the HDPE liner.

During installation of the rain water skirt, if any, it shall be ensured that the HDPE liner is in such a position that ingress of rain water is impossible.

5.7 SPECIAL TANK FOUNDATIONS

5.7.1 Concrete foundation ring under the tank shell

A concrete foundation ring under a tank shell shall not be used except where it is required to provide a downward force to compensate for wind and the uplift caused by internal pressure, e.g., for BHC and BHD (Butt-welded High-pressurized Dome-roof) tanks; see DEP 34.51.01.31-Gen.

Concrete rings often are hard spots under the tank shell and tank bottom periphery plates and this may result in unacceptable differential bottom settlements over very short distances. This could result in cracks in plates through which produce leakage could occur.

If it is established through calculation that this risk exists, concrete slabs shall be used to prevent a sharp transition; see (Appendix 10) for a typical section.

If concrete foundation rings are used, the top of the ring shall be covered with a bitumen layer of at least 5 mm to prevent corrosion.

A stability calculation shall be carried out to determine the number of anchor bolts required.

5.7.2 Concrete rafts

For small tanks, a concrete raft is sometimes used as a foundation. In this case, the top of the concrete raft shall be covered with a layer comprising a mixture of sand and bitumen or a cement stabilised sand bed of at least 50 mm to allow movement of the tank bottom and to prevent corrosion.

The possible total and differential settlements, stress distributions under the slab, and the strength of the slab, taking into account final settlement, shall be determined and incorporated into the design.

5.7.3 Concrete piled foundations

A concrete piled foundation shall be used only when unacceptable settlements are expected.

5.7.4 Redevelopment of areas

There are hazards associated with the construction of new tanks in areas that have previously been used. Some typical aspects that should be investigated include:

- the possibility of differential settlement resulting from varying settlement characteristics due to selected areas being subject to preloading;
- obstructions which require removal such as obsolete foundations or piles;
- the possibility of preferential contamination seepage paths at the location of redundant piles or where vertical sand drains, etc., have been installed.

An investigation shall be required to assess all potential impacts of these and other additional hazards to ensure the resulting design meets the requirements of this DEP.

6. TANK MAINTENANCE

Before jacking up of the tank bottom, the average existing tank pad level shall be measured to determine the (extra) height of the future tank pad, also the orientation of the tank nozzles shall be measure prior to jacking up.

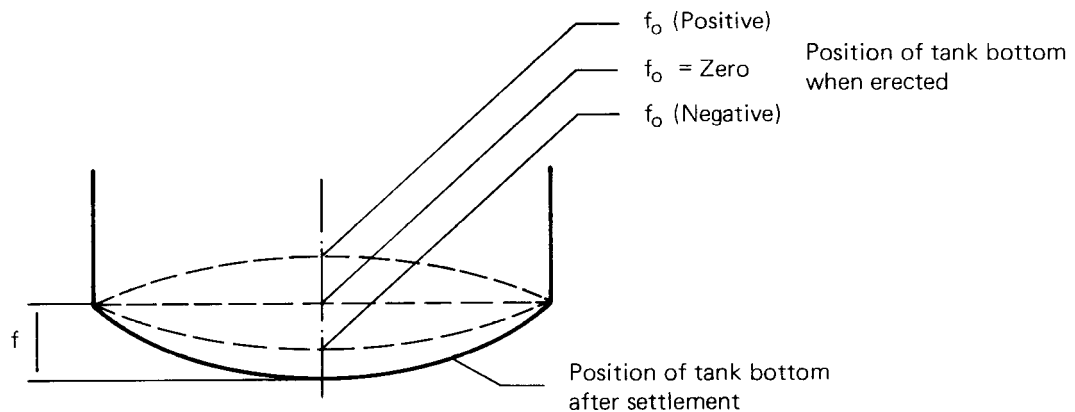
The tank pad shall be corrected if one of the following criteria applies:

- distance between the highest point of capillary groundwater and lowest point of the tank bottom (sump bottom) is less than 100 mm;
- the remaining height of the pad from the lowest point of the tank shell/bottom connection to grade is less than 300 mm;
- maximum allowable uneven settlement (tilting), or height difference of adjacent measuring points of a tank is exceeded;
- permissible angular displacement of the shell/bottom wall connection is exceeded.

Existing tanks shall only be provided with a leak detection system if and when the tank pad is renovated (tank is jacked up). A leak detection system shall only be installed after any existing pollution has been removed from the tank pad.

For tank maintenance, see DEP 70.51.10.11-Gen., from which the following is taken:

Figure 3 Maximum Allowable Sag in the Tank Bottom (After Settlement)



The maximum sag (f) should not exceed the value derived from the following equation:

$$f = \frac{D}{100} \sqrt{\left(\frac{100f_o}{D}\right)^2 + 3.75}$$

where:

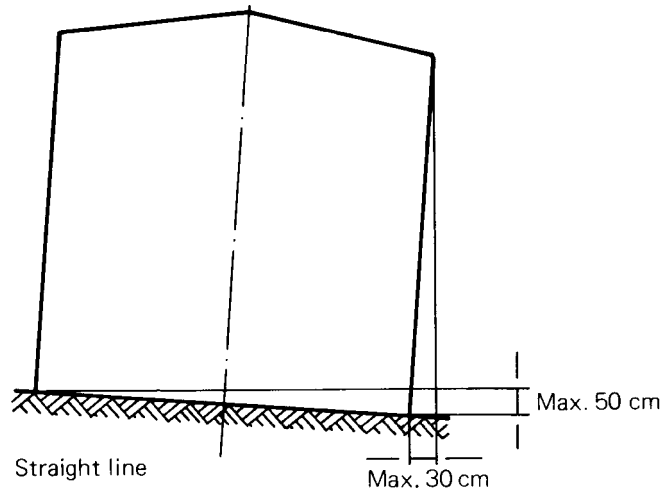
f = maximum allowable sag in tank bottom in m;

D = diameter of tank in m;

f_o = deflection of centre of tank bottom in m. This is in relation to bottom edge at time of tank erection (positive, zero or negative).

Note: It is recommended to use a safety factor of 2 for this formula, in view of possible reduced thickness (corrosion) and distortions.

Figure 4 Maximum Allowable Uneven Settlement (Tilting)



Before these maximum values are reached, the tank should be lifted (i.e. jacked up by a specialist contractor) and the foundation repacked.

It is not always necessary to lift the tank completely. For instance, when a floating roof tank becomes too oval because of partial settlement of the tank wall, it may be sufficient to lift the tank wall only to such a level that the top of the shell is round again.

During the jacking operation, the outer 2 m to 4 m of the tank bottom is often lifted off the tank foundation. This space can be filled with clean dry sand that is either blown in place or manually inserted.

After this exercise, it will be necessary to relevel and resurface the tank shoulder.

For requirements regarding tolerances in tank constructions, acceptability criteria at inspections and allowable deformations after hydrostatic testing, reference is made to DEP 64.51.01.31-Gen.

7. DRAINAGE OF BUNDED AREAS

Drainage of banded areas shall be designed and constructed in accordance with the requirements of DEP 34.14.20.31-Gen.

In areas with high rainfall, frequent draining of compounded drains shall be undertaken to prevent flooding and saturation of bund walls.

In unmanned depots or depots left unattended at weekends where there is no automated control system to drain banded areas, arrangements shall be made to drain the banded area(s) after heavy rainfall. Alternatively, remote electrical level control indicators could be installed to alert personnel when banded areas require draining, or to activate an automatic pumping system.

8. REFERENCES

In this DEP, reference is made to the following publications:

- NOTES:
1. Unless specifically designated by date, the latest edition of each publication shall be used, together with any amendments/supplements/revisions thereto.
 2. The DEPs and most referenced external standards are available to Shell users on the SWW (Shell Wide Web) at <http://sww.shell.com/standards>.

SHELL STANDARDS

Index to DEP publications and standard specifications	DEP 00.00.05.05-Gen.
Index to standard drawings	DEP 00.00.06.06-Gen.
Amended per Circular 60/08	
Definition of temperature, pressure and toxicity levels	DEP 01.00.01.30-Gen.
Minimum requirements for structural design and engineering	DEP 34.00.01.30-Gen.
Site investigations	DEP 34.11.00.10-Gen.
Geotechnical and foundation engineering	DEP 34.11.00.12-Gen.
Roads, paving, surfacing, slope protection and fencing	DEP 34.13.20.31-Gen.
Drainage and primary treatment facilities	DEP 34.14.20.31-Gen.
Standard vertical tanks – selection, design and fabrication	DEP 34.51.01.31-Gen.
Mounded horizontal cylindrical bulk storage vessels for pressurised gases at ambient temperatures	DEP 34.51.11.30-Gen.
Standard vertical tanks – field erection and testing	DEP 64.51.01.31-Gen.
Field inspection, maintenance and repair of vertical steel storage tanks	DEP 70.51.10.11-Gen.

STANDARD DRAWINGS

Amended per Circular 60/08	
General tank foundation without leak detection	S 12.001
Bund wall - typical details	S 12.002
Conceptual tank foundation and tank leak detection and management system	S 12.003
Holding-down bolts for tanks with concrete foundation ring	S 51.046
Typical side drain with sump	S 51.088

AMERICAN STANDARD

Coastal Engineering Manual

Issued by:
US Army Corps of Engineers
Publications Depot
US Government Printing Office,
Washington DC, USA

Design and construction of large, welded, low-pressure storage tanks

API STD 620

Issued by:
American Petroleum Institute
1220 L Street, NW
Washington, DC 20005
USA

Standard practice for classification of soils for engineering purposes (unified soil classification system) ASTM D 2487

Standard practice for description and identification of soils (visual-manual procedure) ASTM D 2488

Standard test method for density of soil and rock in-place at depths below the surface by nuclear methods ASTM D 5195

Issued by:
ASTM International
100 Barr Harbor Drive
West Conshohocken, PA 19428-2959 USA

BRITISH STANDARDS

Methods of test for soils for civil engineering purposes – Part 1 through Part 9 BS 1377

Methods of test for petroleum and its products Part 72 – Determination of viscosity of cutback bitumen BS 2000-72

Specification for manufacture of vertical steel welded non-refrigerated storage tanks with butt-welded shells for the petroleum industry BS 2654

Code of practice for earthworks BS 6031

Code of practice for maritime structures – Part 1 through Part 7 BS 6349

Glossary of building and civil engineering terms - Part 1: General and miscellaneous - Section 1.0 General BS 6100-1.0

Glossary of building and civil engineering terms - Part 2: Civil engineering - Section 2.2: Substructures, earthworks, foundations, tunnels - Subsection 2.2.2: Substructures and foundations BS 6100-2.2.2

Code of practice for design of concrete structures for retaining aqueous liquids BS 8007

Issued by:
British Standards Institution
389 Chiswick High Road
London W4 4AL, UK

OTHER STANDARDS / PUBLICATIONS

Manual on the use of rock in coastal and shoreline engineering CUR/CIRIA 154

Issued by:
CUR (Civieltechnisch Centrum Uitvoering Research and Regelgeving)
[Dutch Organization for Research and Codes in Civil Engineering]

European model code of safe practice in the storage and handling of petroleum products Part II

Issued by:
Applied Science Publishers LTD
Ripple Road,
Barking, Essex, UK

APPENDIX 1 FIELD COMPACTION TRIALS – SUGGESTED PROCEDURE

A test area 20 m long by 15 m wide should be prepared on the actual site of construction from which the topsoil has been removed. The fill material to be used should then be spread over this area in three strips 5 m wide, the depth of the loose material in the strips being varied over the range of thicknesses. Normally, the range of values investigated would be from 150 mm to 450 mm. No adjustment should be made to the moisture content of the fill material that, apart from minor fluctuations, should be in its natural moisture condition.

The test area should then be compacted with the plant it has been decided to use and the mean bulk density, moisture content and dry density of the full depth of each strip should be determined after 2, 4 and 8 passes for all types of plant (except for sheepfoot rollers, for which the density and moisture content determinations should be made after 4, 8 and 16 passes). The densities of the soil should be determined by the core-cutter, sand replacement, gamma radiation or NDP method, whichever is most suitable, and the mean of five determinations should be obtained for each soil condition. Samples of the fill material should be taken and the maximum dry density and optimum moisture content determined (BS 1377).

This procedure should be repeated, if possible, at two other moisture contents, the values suggested being the optimum moisture content given by the BS compaction test (BS 1377) and an intermediate moisture content between this value and the natural moisture content of the soil. If the natural moisture content of the soil is similar to the optimum moisture content, it is suggested that the additional tests be conducted at moisture contents 3 % on either side of this value. These tests should be conducted with fresh fill material laid on an adjacent area from which the topsoil has been removed.

The dry density and moisture content test results should be compared against the maximum dry density that is achieved at the Optimum Moisture Content, to determine the percentage of compaction achieved in the construction trial. This degree of compaction should be compared with the requirements of the specification to ensure compliance.

Soil resistance to penetration can be measured to verify the compaction trials, using Cone Penetration Test (CPT). CPT measurements shall have associated water pressure measurements.

The test procedure has been described in detail, but it can usually be considerably shortened by noting the trend of the early results, and by the experience of the engineers carrying out the tests. These trials take about a week to complete, and could be started as soon as the necessary plant arrives on the site, while the preliminary clearing operations are in progress. The trials can thus be completed before the main construction starts. From the results of the trials and knowledge of the costs of the various procedures, it is possible to determine the most economical procedure to be adopted to obtain the specified degree of compaction, and the subsequent control work can then be finalised.

APPENDIX 2 INDICATIVE COMPACTION VALUES

The following table provides the typical compaction efforts required to achieve a sufficient degree of compaction by means of various equipment on different types of fill material. Due to the inherent variability of fill materials and the non-standard compaction equipment that may be proposed, a construction trial shall be carried out to confirm the proposed method will meet the specific design requirements.

Type of compaction plant	Category	Cohesive soils		Well-graded granular and dry cohesive soils		Uniformly graded material	
		D	N	D	N	D	N
Smooth-wheeled roller	> 2100 kg and ≤ 2700 kg	125	8	125	10	125	10*
	> 2700 kg and ≤ 5400 kg	125	6	125	8	125	8*
	> 5400 kg	150	4	150	8	unsuitable	
Grip roller	> 2700 kg and ≤ 5400 kg	150	10	unsuitable		150	10
	> 5400 kg and ≤ 8000 kg	150	8	125	12	unsuitable	
	> 8000 kg	150	4	150	12	unsuitable	
Tamping roller	> 4000 kg	225	4	150	12	250	4
Pneumatic-tyred roller	Mass per wheel						
	> 1000 kg and ≤ 1500 kg	125	6	unsuitable		150	10*
	> 1500 kg and ≤ 2000 kg	150	5	unsuitable		unsuitable	
	> 2000 kg and ≤ 2500 kg	175	4	125	12	unsuitable	
	> 2500 kg and ≤ 4000 kg	225	4	125	10	unsuitable	
	> 4000 kg and ≤ 6000 kg	300	4	125	10	unsuitable	
	> 6000 kg and ≤ 8000 kg	350	4	150	8	unsuitable	
	> 8000 kg and ≤ 12,000 kg	400	4	150	8	unsuitable	
Vibrating roller	> 12,000 kg	450	4	175	6	unsuitable	
	Mass per metre width of a vibrating roller						
	> 270 kg and ≤ 450 kg	unsuitable		75	16	150	16*
	> 700 kg and ≤ 1300 kg	100	12	125	12	150	6
	> 1300 kg and ≤ 1800 kg	125	8	150	8	200	10*
	> 1800 kg and ≤ 2300 kg	150	4	150	4	225	12*
	> 2300 kg and ≤ 2900 kg	175	4	175	4	250	10*
	> 2900 kg and ≤ 3600 kg	200	4	200	4	275	8*
	> 3600 kg and ≤ 4300 kg	225	4	225	4	300	8*
	> 4300 kg and ≤ 5000 kg	250	4	250	4	300	6*
	> 5000 kg	275	4	275	4	300	4*
Vibrating-plate compactor	Mass per unit area of base plate:						
	> 880 kg and ≤ 1100 kg	unsuitable		unsuitable		75	6
	> 1100 kg and ≤ 1200 kg	unsuitable		75	10	100	6
	> 1200 kg and ≤ 1400 kg	unsuitable		75	6	150	6
	> 1400 kg and ≤ 1800 kg	100	6	125	6	150	4
	> 1800 kg and ≤ 2100 kg	150	6	150	5	200	4
Vibro-tamper	> 2100 kg	200	6	200	5	250	4
	Mass:						
	> 50 kg and ≤ 65 kg	100	3	100	3	150	3
Power rammer	> 65 kg and ≤ 75 kg	125	3	125	3	200	3
	> 75 kg	200	3	150	3	225	3
Power rammer	Mass:						
	100 kg and ≤ 500 kg	150	4	150	6	unsuitable	
Dropping weight compactor	> 500 kg	275	8	275	12	unsuitable	
	Mass of rammer > 500 kg						
	Height of drop:						
	> 1 m up to 2 m	600	4	600	8	450	8
	> 2 m	600	2	600	4	unsuitable	

D = Maximum depth of compacted layer (mm)

N = Minimum number of passes

* = rollers shall be towed by track-laying tractors. Self-propelled rollers are unsuitable.

DEFINITIONS AND REQUIREMENTS ASSOCIATED WITH THE TABLE

- A. The depth of compacted layer is the height by which the embankment is raised by each successive compacted layer.
- B. The number of passes is the number of times that each point on the surface of the layer being compacted has been traversed by the item of compaction plant.
- C. The compactive effort of each compactor is a function of the mass of the machine and the compaction plant. In the table above, the mass per metre width of roll is the total mass of the roller divided by the total roller width. Where a smooth-wheeled roller has more than one axle, the machine shall be assessed on the basis of the axle giving the highest value of mass per metre width.
- D. A tamping roller, for the purposes of this DEP, is a machine with a roll or rolls from which "feet" project. The projected end area of each "foot" shall exceed 0.01 m^2 and the sum of the area of the feet shall exceed 15 % of the surface area of area of the cylinder. The requirements for a tamping roller apply to machines that have 2 rolls in tandem. If only one tamping roll traverses each point on the surface of the layer on any one pass of the machine, the minimum number of passes shall be twice the number given in the table.
- E. (i) For pneumatic-tyred rollers, the mass per wheel is the total mass of the roller divided by the number of wheels.
- (ii) In assessing the number of passes of pneumatic-tyred rollers, the effective width shall be the sum of the widths of the individual wheel tracks together with the sum of the spacing between the wheel tracks provided that each spacing does not exceed 230 mm. Where the spacing between wheel tracks exceeds 230 mm the effective width shall be the sum of the widths of the individual wheel tracks only.
- F. Vibrating rollers are self-propelled or towed smooth-wheeled rollers having means of imparting mechanical vibration to one or more rolls.
- (i) The requirements for vibrating rollers are based on the use of the lowest gear on a self-propelled machine with mechanical transmission and a speed of 1.5 km/h to 2.5 km/h for a towed machine, or a self-propelled machine with hydrostatic transmission. If higher gears or speeds are used an increased number of passes shall be provided in proportion of the increase in speed of travel.
- (ii) Where the mechanical vibration is imparted to two rolls in tandem, the minimum number of passes shall be half the number given in the table above for the appropriate mass per metre width of one vibrating roll. If one roll differs in mass per metre width from the other, the number of passes shall be calculated as for the roll with the smallest value. Alternatively, the machine may be treated as having a single vibrating roll with a mass per metre width equal to that of the roll with the higher value.
- (iii) Vibrating-type rollers operating without vibration will be classified as smooth-wheeled rollers.
- (iv) Vibrating rollers shall be operated with their vibratory mechanism operating only at the frequency of vibration recommended by the Manufacturers. All such rollers shall be equipped or provided with a device automatically indicating the frequency at which the mechanism is operating.
- G. Vibrating-plate compactors are machines having a baseplate to which is attached a source of vibration consisting of one or 2 eccentrically weighted shafts.
- (i) The mass per unit area of baseplate of a vibrating-plate compactor is calculated by its working condition by its area in contact with compacted soil.
- (ii) Vibrating-plate compactors shall be operated at the frequency of vibration recommended by the Manufacturers. They shall normally be operated at travelling speeds of less than 1 km/h but if higher speeds are necessary the number of passes shall be increased in proportion to the increase in speed of travel.
- H. Vibro-tampers are machines in which an engine-driven reciprocating mechanism acts on a spring system through which oscillations are set up in a baseplate.

- I. Power rammers are machines that are actuated by explosions in an internal combustion cylinder, each explosion being controlled manually by the operator.
- J. In the case of power rammers and dropping-weight compactors, one pass will be considered as made when the compacting shoe has made one strike on the area in question.

APPENDIX 3 MIXTURES OF SAND AND BITUMEN FOR TANK FOUNDATIONS

The sand shall be clean and have a silt content of less than 5 %. The amount of material passing the 200 mesh (.74) sieve should be 3 % to 5 % (this can usually be obtained by blending sands from different sources if the preferred source is unsuitable). It is not possible to give precise grading limits as satisfactory mixes can be made from a large variety of sands of widely different grading.

Cutbacks or penetration grade bitumen may be used. The following table lists potentially suitable grades together with recommended ranges for mixing temperatures. These mixing temperature ranges should be strictly complied with to ensure that good coating of the minerals is obtained and that the bitumen is not damaged by excessive heat.

Grade	Recommended mixing temperature ranges, (°C)	
	Minimum	Maximum
Shelmac 50/100	65	95
Shelmac 150/200	80	95
Shelmac 200/300	85	105
Shelmac 300/400	95	110
Shelmac 400/500	100	115
Shelmac 500/700	110	120
Shelmac S.125	105	135
Shelmac SRO	70	100
MC 800	80	115
MC 3000	95	120
MC 3	65	95
MC 4	80	100
MC 5	100	120
Mexphalte 80/100	135	160
Mexphalte 60/70	145	170

The most suitable bitumen content is best assessed through mixing trials using the equipment available. The assessment should include the workability, visual and handling characteristics of the mix. A guide can be obtained from the grading of the sand by using a method based on the specific surface area of the particles.

A conventional bitumen asphalt mixer of either the batch or continuous type, having a twin-shaft pugmill, is generally necessary. At remote sites, this may not be possible and it may sometimes be necessary to mix by hand, although this is only possible when using low viscosity cutbacks. The sand must be dried on a steel plate over a fire and the bitumen heated prior to mixing by turning over with shovels.

The mix can be spread by hand, using wooden side forms and screeds but for large-scale work where equipment is available, a grader or asphalt paver may be more appropriate. When spreading by hand, the mix should be tipped on to shovelling plates outside the area on which it is to be placed and then moved from the heap and placed in position for screeding. The total compacted thickness required will depend on the state of the subgrade: if it is uniform and well-compacted a total compacted thickness of 50 mm of the bitumen and sand mix will normally be sufficient but if the subgrade is poorly compacted, it may be desirable to place two 50 mm layers of mix.

Sand mixes compact easily and a few passes of light pneumatic tyred rollers or tractors provide effective compaction. The marks caused by tractor tyres can be ironed out with passes of a steel tyred roller or vibrating-plate compactor that usually provide an adequate finish.

For **hot tanks** use an oil and sand mix, 50 l per m³ sand (oil: Shell Carnea or equivalent).

APPENDIX 4 WET SAND MIX FOR TANK FOUNDATION SHOULDERS AND BUND WALLS

The majority of sands are suitable but they shall not have silt contents of more than 3 %. It is not appropriate to give any definite limits to the grading required since satisfactory mixes can be made with a large variety of sands of widely different gradings. It is, however, desirable that there should be 3 % to 10 % of material passing the 200-mesh sieve.

The bitumen is a specially prepared kerosine cutback known as Shelmac SRO (Special Road Oil) with a viscosity of 40 s to 50 s STV (standard tar viscosity)* at 25 °C, containing an additive. For use in the tropics, a grade made with harder basic bitumen, known as Tropical SRO, should be used. SRO can be made available on special order from some Shell refineries.

Note*: For further information, see BS 2000-72.

In order to coat the wet sand an activator is required in conjunction with the additive in the SRO. The most effective activator is hydrated lime with a calcium hydroxide content of 90 %. Although less is required for the chemical reaction, 2 %, based on the total mix, is used to ensure good dispersion in the sand. If hydrated lime is not available, Portland cement may often be used in which case 3 % to 4 % is required, primarily due to the greater density of cement. The activator is dispersed in the sand before mixing with the SRO. Larger quantities of activator should not be used, as it is not required for the chemical reactions; it will require additional SRO because of the high specific surface area of the activator and may result in a rather brittle mix of reduced durability.

Water is an essential component in the process and even if the sand is damp it will usually be necessary to add additional water to bring the total water content of the sand to at least 5 %.

The minimum SRO content is determined by a simple cohesive test that is described as follows. A mix is made using dry sand, 4 % SRO, 2 % hydrated lime and 10 % water. The mix may be prepared by hand mixing. The mix is introduced into a wide mouthed glass jar to give a loose layer of about 15 mm and covered to a depth of about 50 mm with water. The mix is then broken up by stirring and then compacted under water using a tamping rod with a base of about 20 mm diameter. If the mix compacts well and resists disintegration when the jar is rotated to swirl the water, the mix is considered to be cohesive. Other mixes should then be prepared with increasing or decreasing quantities of SRO, usually in steps of ½ %, to determine the minimum quantity required to obtain a cohesive mix. A suitable working SRO content is considered to be the minimum plus 1½ %. If the working SRO content is less than 4 %, as may be the case with sands lacking fines, it may be desirable to increase the fines content.

All percentages are in terms of weight.

In general, a mix made with the working SRO content will be satisfactory but there are conditions where the mix will not harden after placement, due to impermeability of the mix, which prevents evaporation of the water and the solvent from the cutback.

This is most likely to occur with fine-graded or 'dirty' sands which require a high SRO content. A small sample should therefore be prepared by rolling some mix in a tray. As the mix is rolled out excess water should be expelled if the mix is sufficiently permeable. If water is not expelled, it is likely that the mix is impermeable and should thus be rejected.

The mix is placed and compacted in the same manner as for dry sand bitumen mixes.

APPENDIX 5 TYPICAL CPT VALUES TO VERIFY COMPACTION AND QUALITY OF TANK FOUNDATIONS

The minimum required number of CPTs (Cone Penetration Test) is:

- 5 CPTs for tanks up to 12 m diameter;
- 9 CPTs for tanks above 12 m and up to 40 m diameter;
- 13 CPTs for tanks larger than 40 m diameter.

Tank foundations are normally constructed of sand and this is compacted to such a degree that the tank foundation meets the stability and settlement requirements.

The degree of compaction can be reliably checked with the aid of CPTs. Examples of minimum required CPT values for various cases are listed below.

Diameter of storage tank (m)	12	12	24	36	45
Height of storage tank (m)	10	15	16	20	24
Shoulder width of tank foundation (m)	0.7	1	1	1.3	1.5
Safety factor against edge sliding	1.4	1.4	1.4	1.4	1.4
Table min. CPT values	1	2	3	4	5

Ground water level: 1.5 m below surface of tank foundation.

TABLE 1

Depth underneath top tank foundation (m)	Minimum CPT value (MPa)
0.25	4.0
0.50	5.5
0.75	6.0
1.00	6.5
1.50	7.0
2.00	7.5

TABLE 2

Depth underneath top tank foundation (m)	Minimum CPT value (MPa)
0.25	5.0
0.50	6.5
0.75	7.5
1.00	8.0
1.50	8.5
2.00	9.0

TABLE 3

Depth underneath top tank foundation (m)	Minimum CPT value (MPa)
0.25	5.5
0.50	7.0
0.75	8.0
1.00	8.5
1.50	9.0
2.00	9.5

TABLE 4

Depth underneath top tank foundation (m)	Minimum CPT value (MPa)
0.25	6.5
0.50	8.0
0.75	8.5
1.00	9.0
1.50	9.5
2.00	10.0

TABLE 5

Depth underneath top tank foundation (m)	Minimum CPT value (MPa)
0.25	7.0
0.50	8.5
0.75	9.0
1.00	9.5
1.50	10.0
2.00	10.5

The penetration depth shall extend as a minimum to the larger of the following values:

- 5 m below the surface of the tank foundation;
- the full extent of the substrata which has been subject to ground improvement.

This is to ensure the engineering properties of the subsoil beneath the tank foundation meets the minimum values adopted in the design. Should the verification test fail to meet the required investigation depth, the reason for refusal shall be determined and proposals for corrective action shall be submitted for approval to the Principal.

Should the recorded CPT values be below the minimum values presented in Tables 1 to 5 above and, in the case of the substrata, below the assumed design values, the reason for the reduced values shall be determined and proposals for corrective action shall be submitted for approval to the Principal.

APPENDIX 6 HYDROSTATIC TESTING PROCEDURE

1. BASIC DATA

Filling height = H (m)

2. RECOMMENDATIONS

- a. Storage tanks placed on clayey, silty or sandy subsoil should be filled in four stages. Duration and interval between the various stages depend on the subsoil conditions and the settlement behaviour during hydrostatic testing.
- b. Soil data as well as settlement and stability calculations must be available before the tank is tested, and the calculated safety factors must meet the requirements.
- c. The settlement behaviour during and after hydrostatic testing must be monitored.

3. TESTING RATE

The variables referenced in the following section are listed in the table below. During and on completion of every stage, the actual data obtained through monitoring should be compared to that predicted. At any time during the test, should the data indicate that the tank, tank foundation or subsoil is performing in an adverse manner and not as predicted, the test should be terminated and the tank restored to a safe condition. The reason for the apparent anomaly should then be established and a proposal for rectifying the situation shall be submitted to the Principal for approval.

Stage 1

Filling of the tank to 0.5 H . The rate of filling should not exceed “a” metres per day. The total filling per day should be completed in a period of not less than 3 hours. During the ensuing balance of hours in the day's total of 24, the foundation will be allowed to equilibrate and consolidate and this process will be monitored. A minimum monitoring period of 6 hours is considered to be necessary.

At the end of Stage 1, there will be a further X-hour monitoring period. The actual rate of settlement at the end of the monitoring period should be compared with that predicted and if acceptably low, the next stage of the test can commence.

Stage 2

Filling of the tank from 0.5 H to 0.67 H . The rate of filling should not exceed “b” metres per day. The total filling time and equilibration and monitoring period per day should be as for stage 1. A minimum equilibration and monitoring period of 8 hours is considered to be necessary.

At the end of Stage 2, there should be a Y-hour monitoring period. The actual rate of settlement at the end of the monitoring period should be compared with that predicted and if acceptably low, the next stage of the test can commence.

Stage 3

Filling of the tank from 0.67 H to 0.83 H . The rate of filling should not exceed “c” metres per day. The total filling time and equilibration and monitoring period per day should be as for Stage 1. A minimum equilibration and monitoring period of 8 hours is considered to be necessary.

At the end of Stage 3, there should be a Z-hour monitoring period. The actual rate of settlement at the end of the monitoring period should be compared with that predicted and if acceptably low, the next stage of the test can commence.

Stage 4

Filling of the tank from 0.83 H to 1.00 H . The rate of filling should not exceed “d” metres per day. The total filling time and equilibration and monitoring period per day should be as for Stage 1. A minimum equilibration and monitoring period of 10 hours is considered to be necessary.

At the end of Stage 4, the tank must be left fully filled and monitored for at least 4 more days. The actual rate of settlement at the end of the monitoring period should be compared with that predicted and if acceptably low the test can be considered complete.

The daily filling rates and the duration of the various equilibration and monitoring periods depend on the subsoil conditions, and the following values are considered appropriate in most circumstances:

Soil conditions underneath tank foundation	a	b	C	d	X	Y	Z
	(max. value in m)				(min. value in hours)		
Soft clay or silt	2	1	0.75	0.5	48	48	48
Stiff clay	2	1.25	0.75	0.5	36	36	48
Clayey, silty sand	2.5	1.25	0.75	0.5	24	36	48
Sand	2.5	1.5	1	0.75	12	12	24
Weathered rock, cemented soil	-	-	-	-	-	12	48
Fresh rock	-	-	-	-	-	-	24

4. MEASUREMENT OF SETTLEMENTS DURING TESTING

Measurement of the tank shell settlements should be taken from at least 8 points, evenly distributed around the wall at a distance of up to 12 m from the wall. During the period of filling which is to take at least 3 hours in a given 24-hour period, the following daily schedule of measurements could be used:

- prior to filling;
- just after filling;
- 2 hours after filling (optional, depending on the results);
- 5 hours after filling.

In addition, during each equilibration period between the various stages and whilst the tank remains completely full in stage 4 and during the draining of the tank, 3 measurements should be taken daily. However, if the tank foundation is on rock (including weathered rock and cemented soil), the measurements may be limited to (a) prior to hydrostatic testing, (b) during the equilibration period between Stages 3 and 4 and (c) at the end of Stage 4.

5. SETTLEMENT BEHAVIOUR DURING TESTING

A. General

The measured daily settlement rate during the equilibration and monitoring period between the various stages and during the period after the tank is filled (i.e. after completion of stage 4) must diminish with time (e.g. the settlement rate of the second day must be less than that of the first day).

In the event of a static or increased settlement rate during the above-mentioned equilibration and monitoring period between the various stages, and during the period just after filling is completed (Stage 4), there is a danger of instability. In such a case, the Principal shall be informed, the tank shall be (partly) emptied to achieve a stable and safe condition and a geotechnical engineer and a tank specialist should be consulted.

B. Floating roof tanks

Floating roofs are equipped with rim seals that can compensate the out-of-roundness of the tank shell. Out-of-roundness of the tank shell may occur due to:

- a. Limits in accuracy of construction. A tank shell could easily show an out-of-roundness up to approximately 50 mm (depending on diameter) due to practical limits of accuracy during construction.
- b. Out of roundness caused by **uneven soil settlement**. In one direction the shell will increase in diameter, in the other direction it will decrease in diameter.

When the ultimate outward movement of the seal is reached, a gap opens between the seal plates and the shell at some locations. At other locations the ultimate inward movement of the seal may be reached, which may result in the roof becoming stuck so that the roof seal is damaged.

6. EVALUATION OF TANK SETTLING CHARACTERISTICS

Compliance with the above recommended procedures and specifications will theoretically ensure that some of the maximum overall tank settlement will occur during the hydrostatic testing period. Extending the period of testing beyond that scheduled above will increase the percentage of total settlement achieved.

It should be noted that observations regarding the behaviour in the rate of settlement and the total settlements achieved during the period of hydrostatic testing will be useful in making a more precise evaluation of future settlements that can be expected to occur during the operational period.

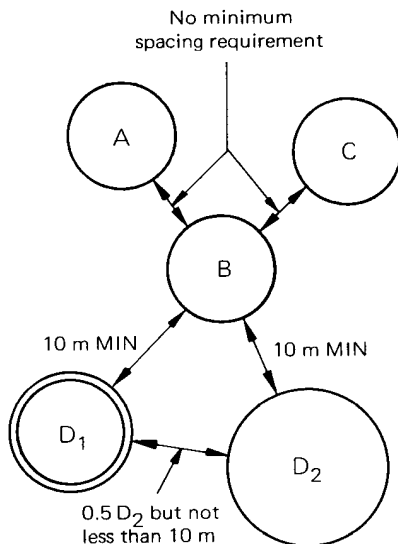
7. FILLING RATE AFTER HYDROSTATIC TESTING

In the event of poor results of the hydrostatic testing, some limitations on the filling rate during the first few operational fillings may be stipulated. A geotechnical or foundation engineer should be consulted.

APPENDIX 7 TYPICAL TANK FARM LAYOUTS WITH SAFETY DISTANCES

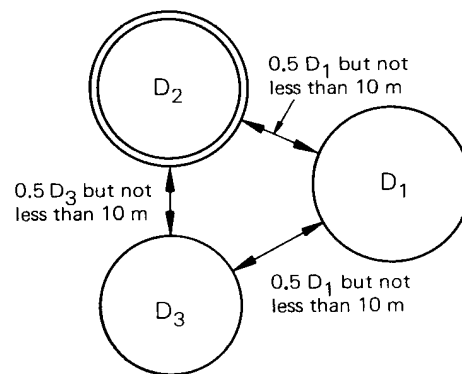
TANKS FOR PETROLEUM STOCKS OF CLASSES I, II AND III (2)

Inter-tank spacings between
small and larger tanks



Tanks A, B, C are fixed or floating roof small tanks (less than 10 m diameter) with a total capacity of less than 8000 m³; no inter-tank spacing requirements other than for construction/operation/maintenance convenience.
Tanks D_1 and D_2 are tanks with diameters greater than 10 m, and with diameter of D_2 greater than D_1 .

Inter-tank spacings
for fixed and floating
roof tanks (> 10 m Ø).



Fixed and floating roof tanks within the same compound, D_1 greater than D_2 , D_2 equal to D_3 .

KEY

- Fixed roof
- Floating roof

NOTE: Classification of these materials, based on closed cup flash points, is as follows:

Class I – Liquids which have flash points below 21 °C;

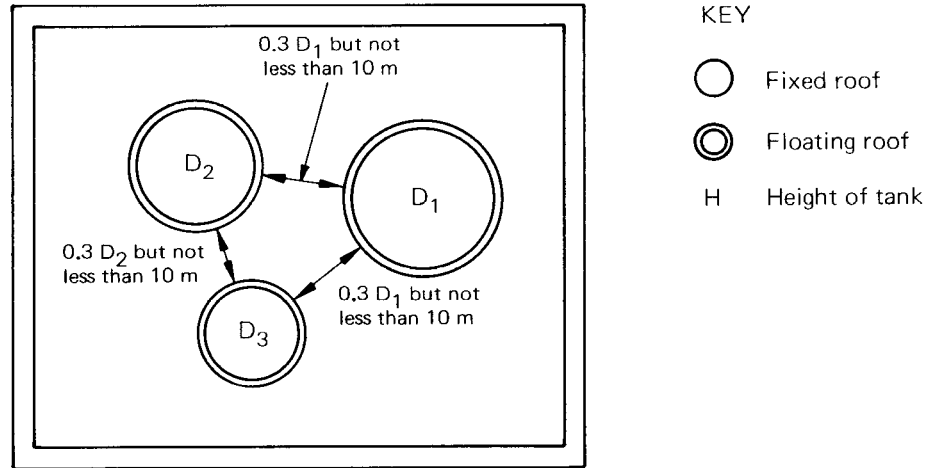
Class II (1) – Liquids which have flash points from 21 °C up to and including 55 °C, handled below flash point;

Class II (2) – Liquids which have flash points from 21 °C up to and including 55 °C, handled at or above flash point;

Class III (2) – Liquids which have flash points above 55 °C up to an including 100 °C, handled at or above flash point.

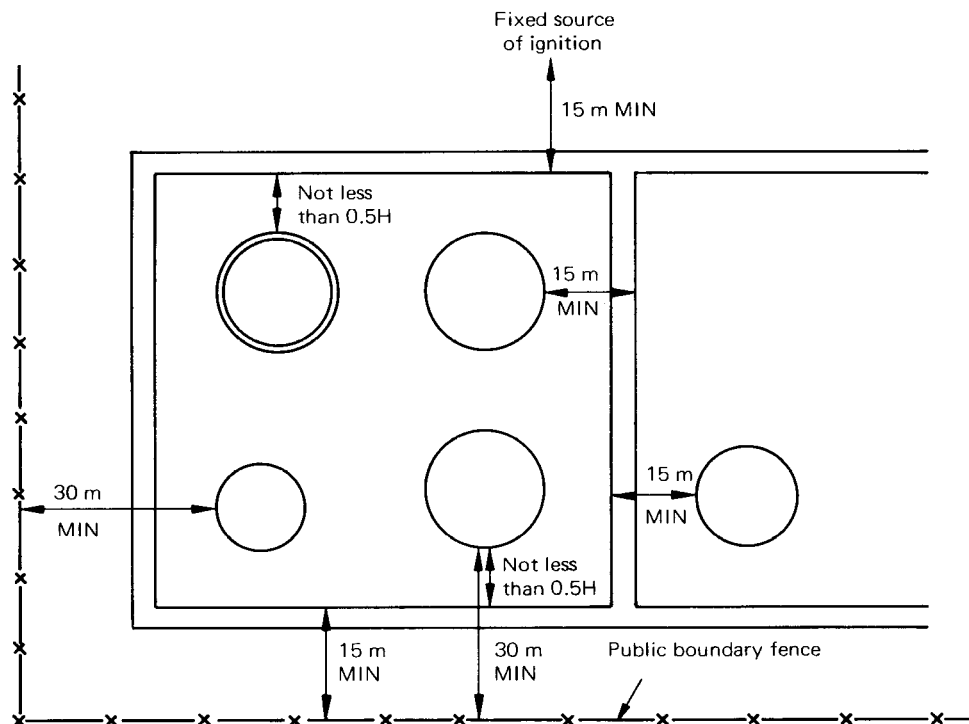
LAYOUT SKETCHES OF TANK FARMS WITH SAFETY DISTANCES
TANKS FOR PETROLEUM STOCKS OF CLASSES I, II AND III (2) (cont'd)

Inter-tank spacings for floating roof tanks ($> 10 \text{ m } \varnothing$).



Floating roof tanks of diameter D_1 D_2 D_3 greater than 10 m within the same compound.
 D_1 greater than D_2 and D_2 greater than D_3 .

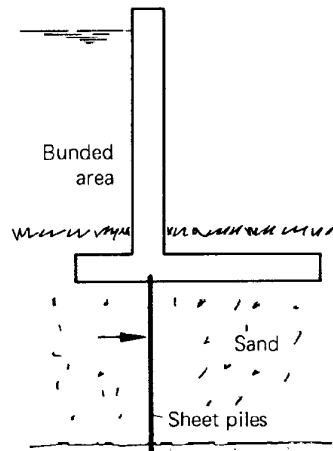
Tank and compound wall distances from typical features



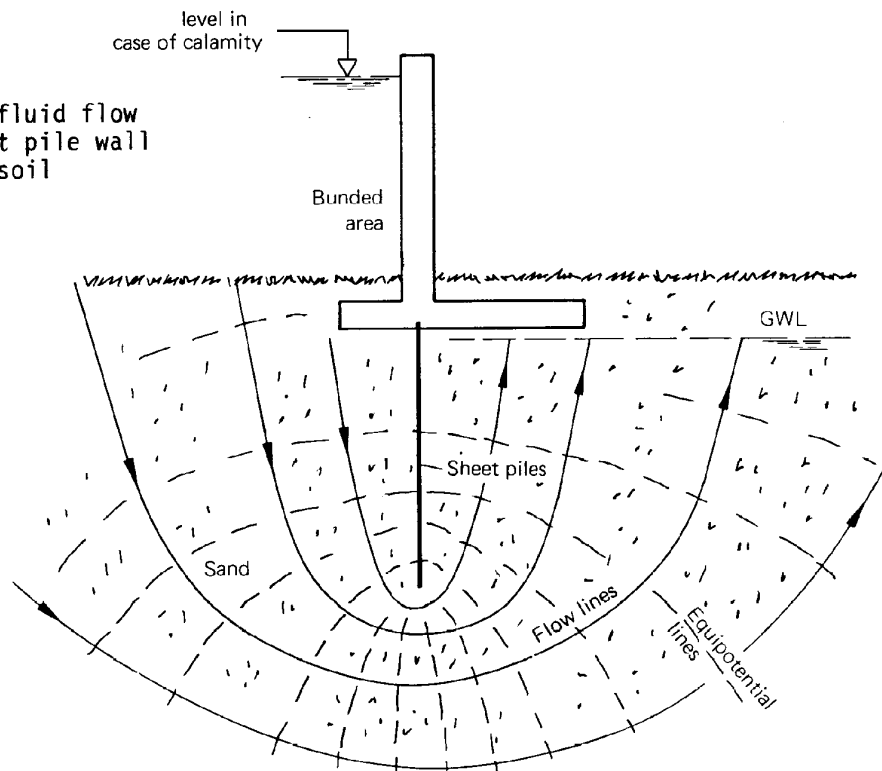
Note: The layout and distances as indicated in the above figures may differ from those indicated in the European Model Code.

APPENDIX 8 TYPICAL EXAMPLES OF RETAINING WALLS

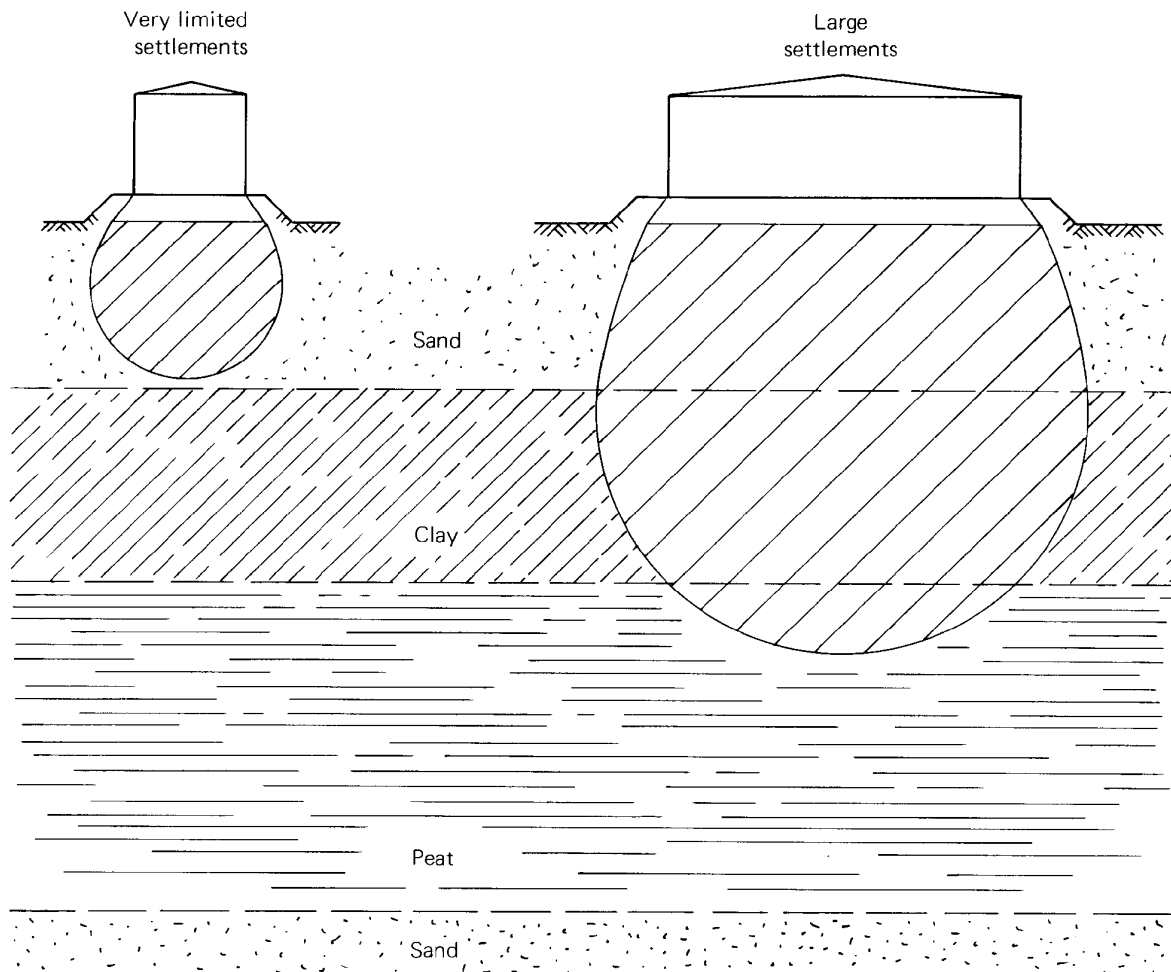
- a) No fluid flow possible
 in a short time
 (couple of days)



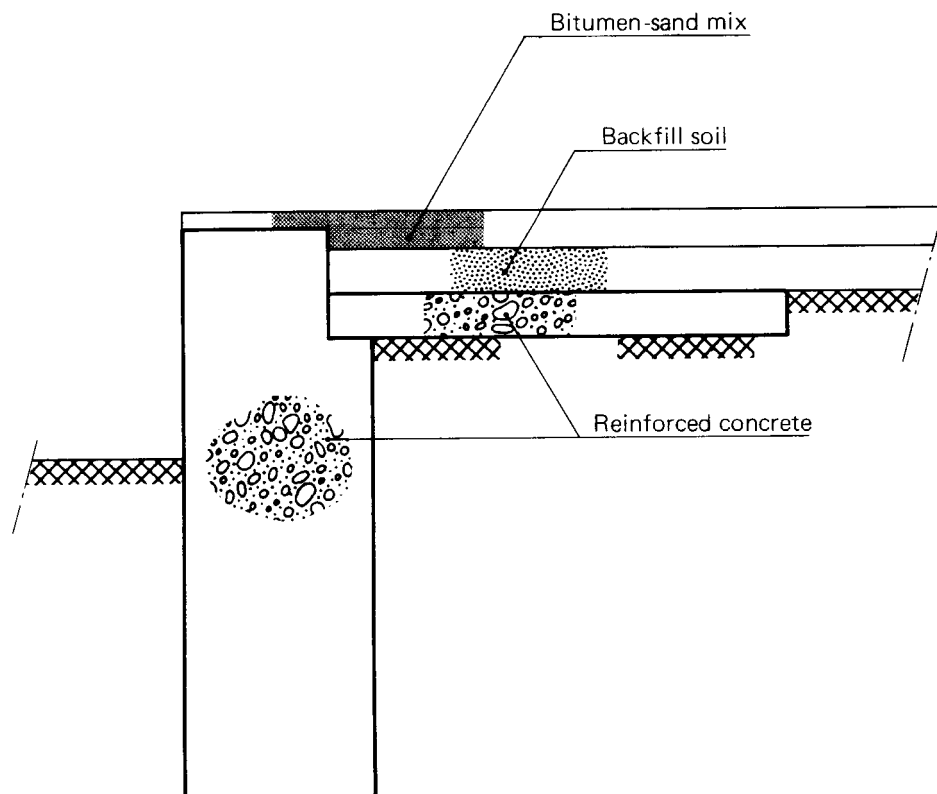
- b) Flow net of fluid flow
 under a sheet pile wall
 in pervious soil



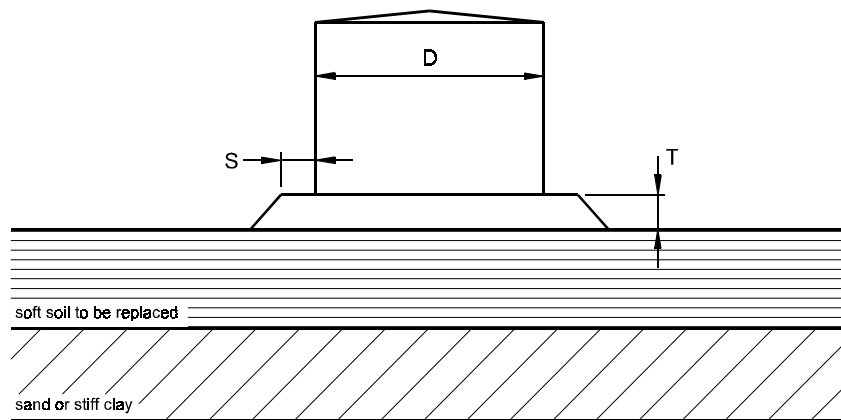
APPENDIX 9 EFFECT OF TANK DIMENSIONS ON SOIL SETTLEMENT



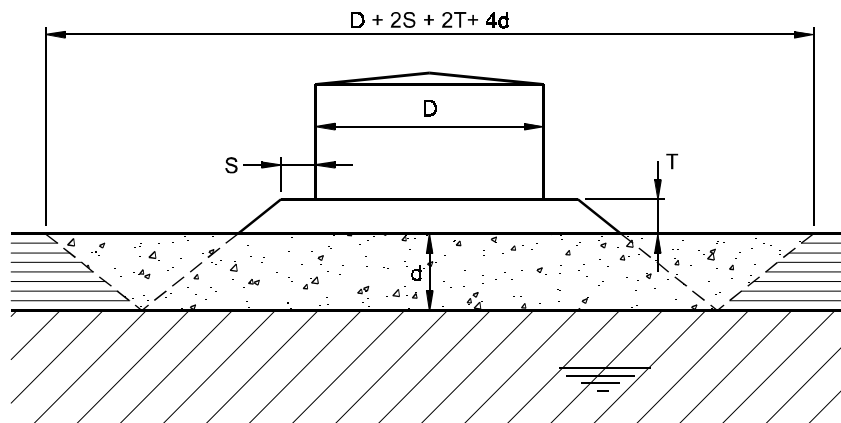
APPENDIX 10 TYPICAL RING FOUNDATION WITH CONCRETE SLAB



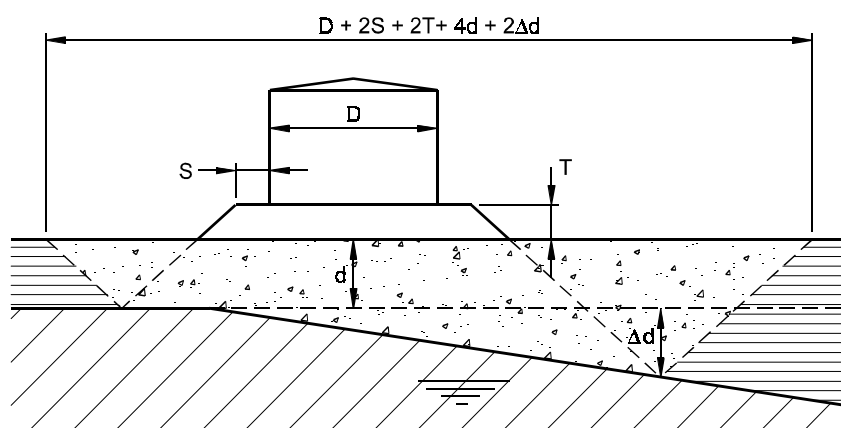
APPENDIX 11 SOIL IMPROVEMENT BY GROUND REPLACEMENT



Section of planned tank location



Circular excavation



Elliptical excavation

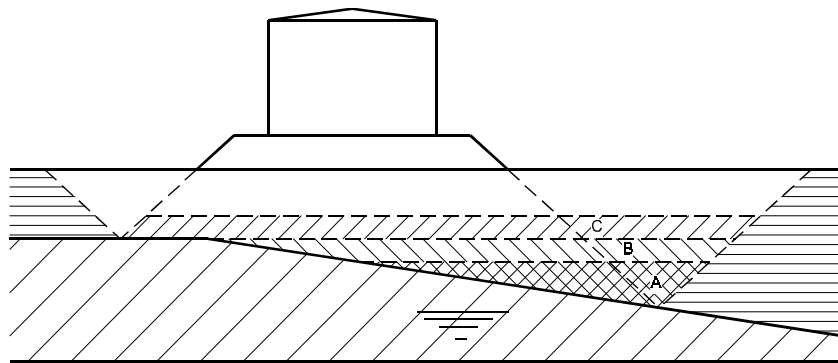
In all cases: Groundwater shall be at least 0.5 m below lowest point of excavation.

After excavation of unsuitable material:

1. Confirm **all** soft soils have been removed; if not, remove soft materials;
2. If subsoil is sand, the exposed excavated surface shall be compacted to achieve a CPT cone resistance of at least 5 MPa or 95 % Proctor.

After the base of the excavation has been adequately prepared, suitable fill material shall be placed and compacted in layers of a thickness appropriate for the compaction equipment employed. The fill material shall be placed at its optimum moisture content and the design strength or density (5 MPa or 95 % Proctor) shall be achieved.

In cases of uneven excavation depth:



- A. first layer of replacement soil.
- B. second layer of replacement soil.
- C. third layer of replacement soil.

Thickness of layers A, B, C etc shall be appropriate for the compaction equipment employed.

This process continues until ground level or the bunded area elevation has been reached, at which point, the construction of the tank foundation can commence.

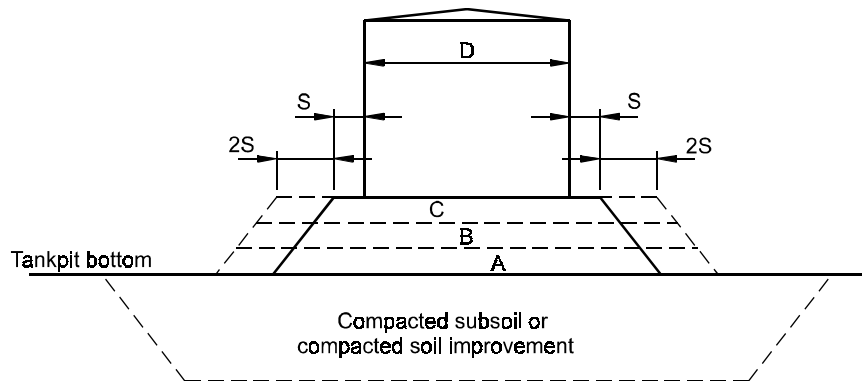
Specifications for sand to be used in soil improvement are given in (Appendix 13).

APPENDIX 12 EXAMPLES OF TANK FOUNDATION CONSTRUCTION

Specifications for crushed stone (20/40) and crushed stone (2/6) are given in (Appendix 13).

Specifications for materials for surface finishing of tank foundations are given in (Appendix 13).

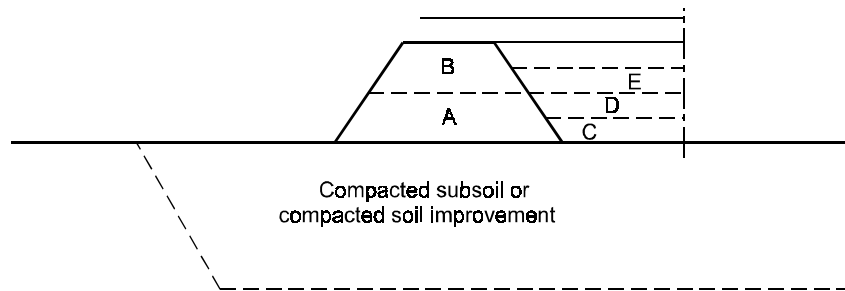
TANK FOUNDATION COMPRISING COMPACTED SAND



1. The first sand layer (A) shall be compacted to achieve a minimum CPT cone resistance of 5 MPa or 98 % Proctor. The thickness of the compacted layer shall not exceed 0.3 m. The sand shall be placed and compacted at Optimum Moisture Content ± 2 %.
2. The placement and compaction of the layers shall be calculated to ensure that the width of the last layer which forms the top surface of the tank foundation shall be $D+6S$. This oversized tank foundation shall be maintained during compaction and construction and is required to ensure the sand which shall form the final tank foundation shall remain protected and hence sufficiently dense.
3. Continue with additional layers (indicated in the figure above as B & C) until the top of the tank foundation or level of the membrane if applicable is reached.
4. Maintain larger diameter of tank foundation during construction works.
5. Remove extra diameter of tank foundation just before hydrostatic testing and shape and protect the outside tank foundation.

Specifications for sand to be used in tank foundation construction are given in (Appendix 13).

CRUSHED STONE RING AND SAND TANK FOUNDATION



Construction of outer crushed stone ring.

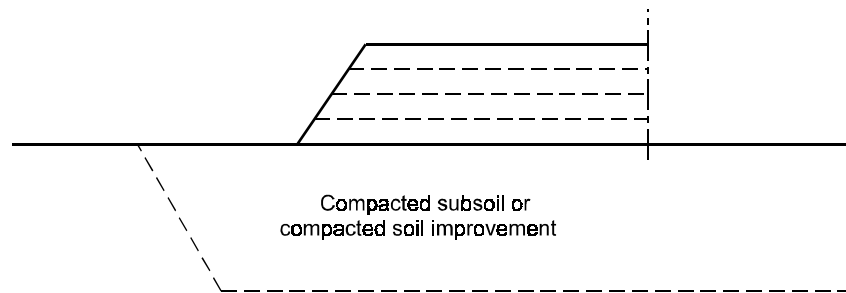
1. Place a permeable geo-textile under the footprint of the proposed crushed stone ring.
2. Place the first layer (A) of crushed stone (20/40) in a maximum thickness of 0.5 m. Spread and compact using a vibrating plate or vibrating roller of approximately 1000 kg. Two passes shall be executed as a minimum.
3. Continue with additional layers (indicated in the figure above as B) until the top of the tank foundation is reached.
4. The crushed stone ring (20/40) shall be finished with a finer crushed stone layer (2/6) on which the tank foundation surface finishing layer shall be placed.

Construction of inner Sand foundation.

This can commence on completion of the outer crushed stone ring and the operation shall be performed as described below:

1. Place a permeable geotextile over the slope of the inside of the crushed rock ring. This is required to prevent migration of sand into the crushed rock ring and thereby prevent associated settlements.
2. The first sand layer (indicated in the figure above as C) shall be compacted with a plate vibrator to achieve a minimum strength or density of 5 MPa or 98 % Proctor. The thickness of the compacted layer shall not exceed 0.3 m. The sand shall be placed and compacted at Optimum Moisture Content ± 2 %.
3. Continue with additional layers (indicated in the figure above as D & E) until the top of the tank foundation or level of the membrane if applicable is reached.
4. Special attention shall be paid to the compaction of the sand close to the stone ring. A minimum of 6 passes with the vibrating machine shall be carried out in the vicinity of the stone ring.

CRUSHED STONE TANK FOUNDATION

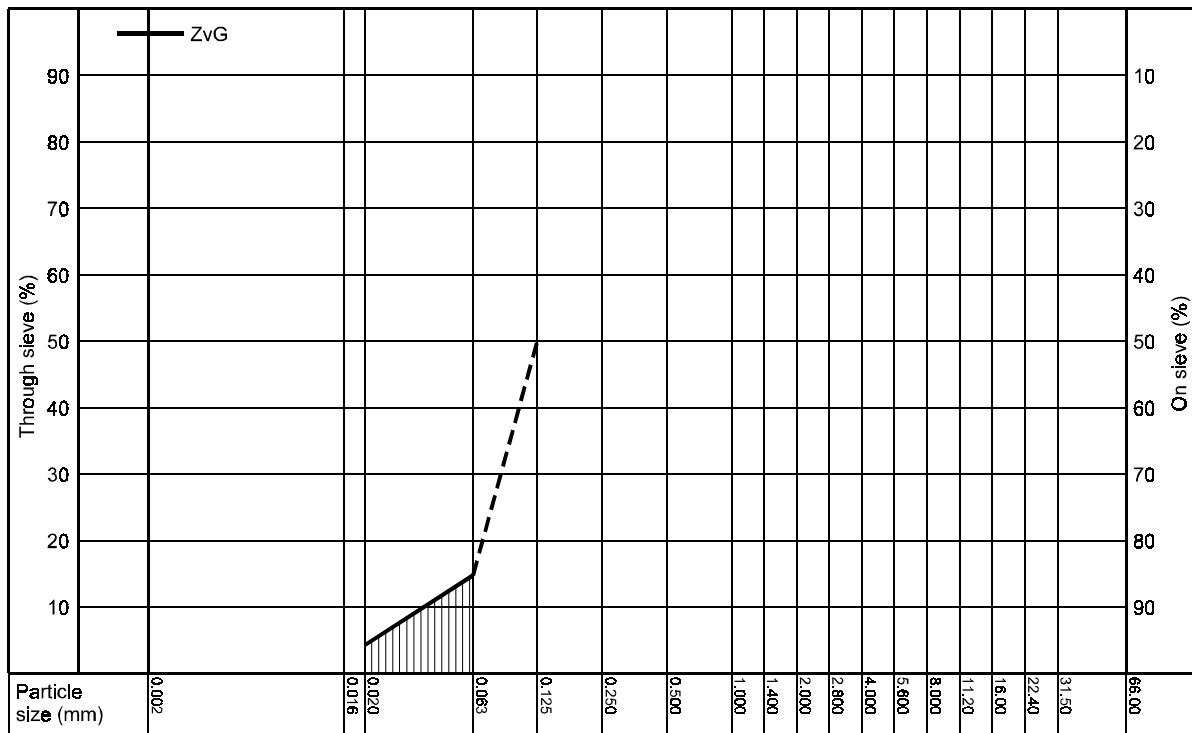


1. Place the first layer of crushed stone (20/40) in a maximum thickness of 0.5 m. Spread and compact using a vibrating plate or vibrating roller of approximately 1000 kg. Two passes shall be executed as a minimum.
2. Continue with additional layers until the top of the tank foundation is reached.
3. The shoulder of the crushed stone (20/40) tank foundation shall be finished with a finer crushed stone layer (2/6) on which the tank foundation surface finishing layer shall be placed.

APPENDIX 13 MATERIALS

Sand for soil replacement

Sample Details			
Borehole No:	Sample No:	Sample Type:	Depth:

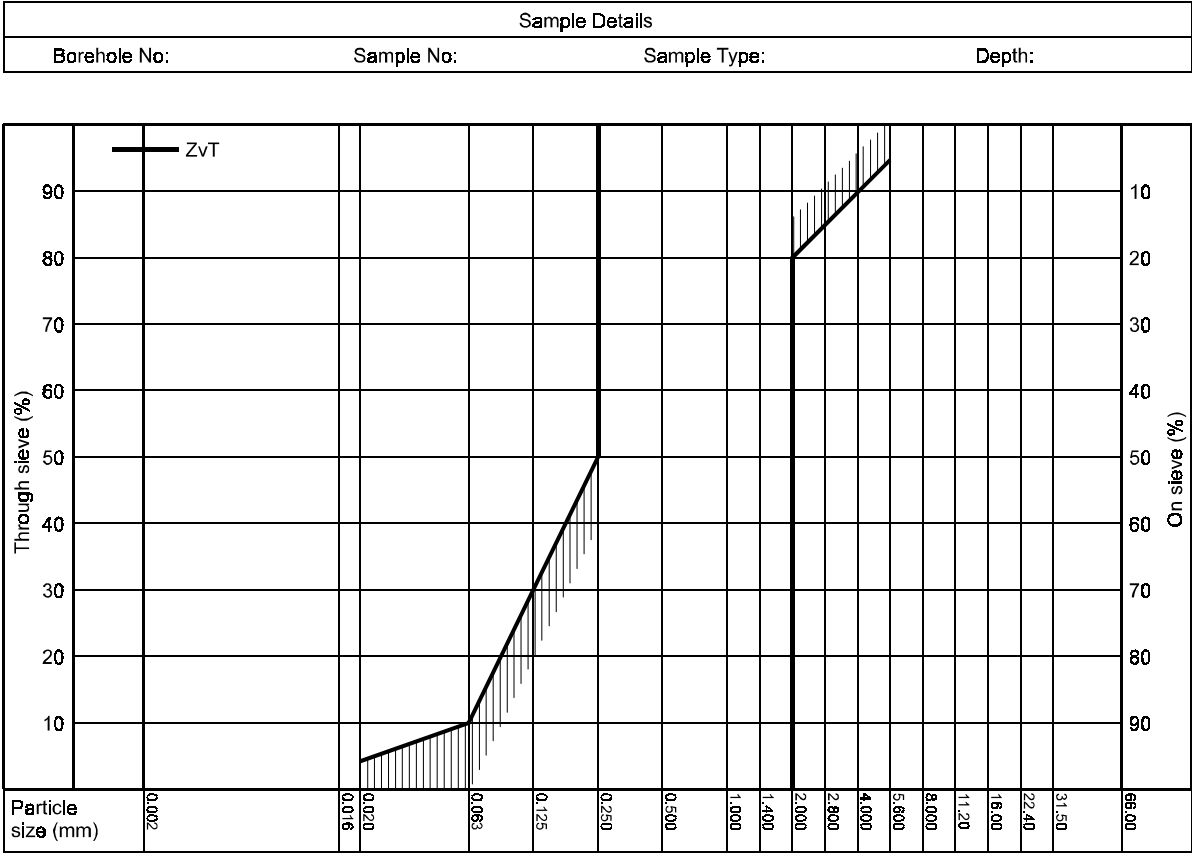


Lutum fract.	Silt fraction	Sand fraction	Gravel fraction

Mineral sand for soil improvement consists of:

organic material	$\leq 3 \text{ \% (m/m)}$
through sieve $< 20 \text{ }\mu\text{m}$	$\leq 5 \text{ \%}$
through sieve $63 \text{ }\mu\text{m}$	$\leq 15 \text{ \%}$
$150 < M_{50}$	$\leq 420 \text{ }\mu\text{m}$
$1.8 < D_{60}/D_{10}$	≤ 3

Sand for tank foundation



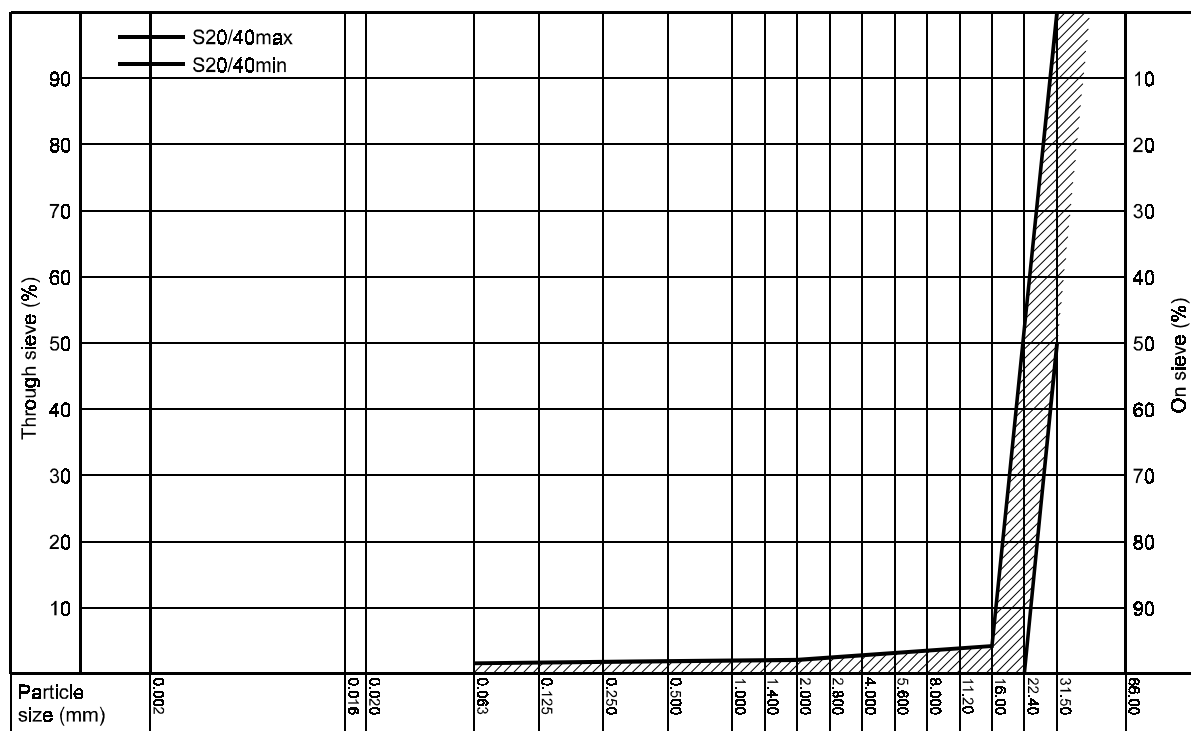
Lutum fract.	Silt fraction	Sand fraction	Gravel fraction

Mineral sand for soil improvement consists of:

- organic material
- ≤ 3 % (m/m)
- through sieve < 20 μm
- ≤ 3 %
- through sieve 63 μm
- ≤ 10 %
- through sieve 250 μm
- ≤ 50 %
- on sieve 2 mm
- ≤ 20 %
- on sieve 5.6 mm
- ≤ 5 %
- M₅₀
- ≥ 250 μm
- 1.8 < D₆₀/D₁₀ ≤ 3

Crushed stone for crushed stone ring
(Crushed stone 20/40)

Sample Details			
Borehole No:	Sample No:	Sample Type:	Depth:

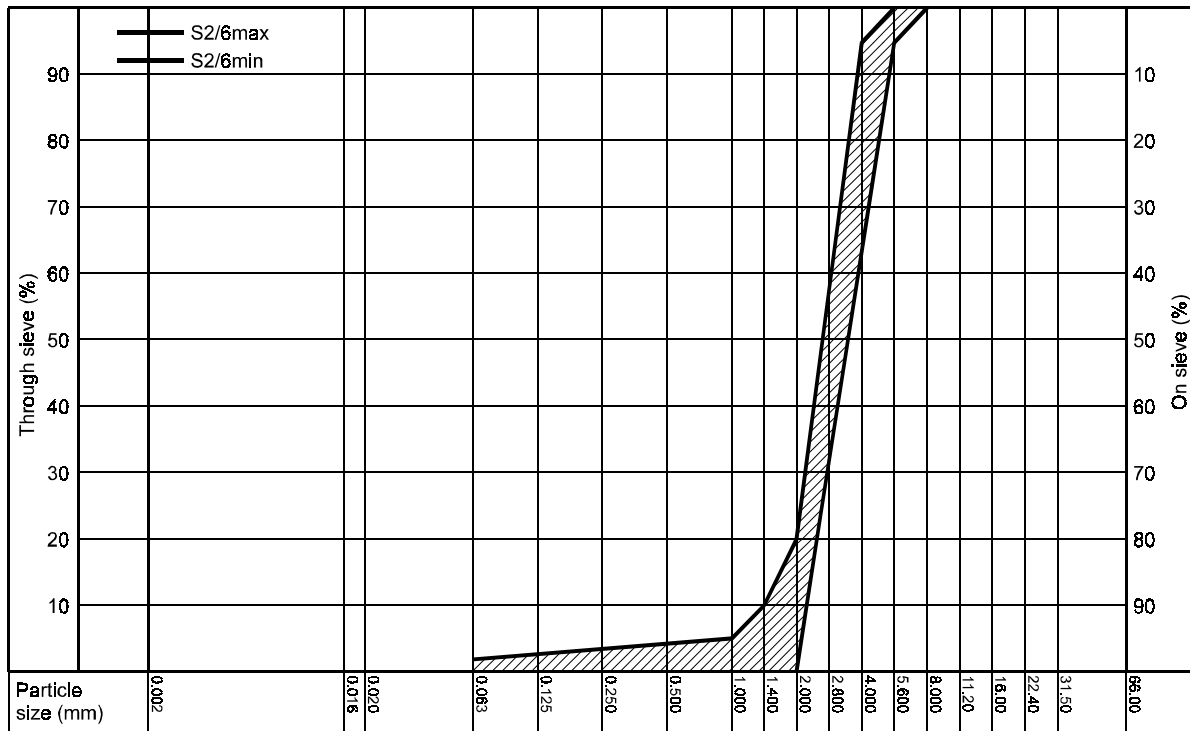


Lutum fract.	Silt fraction	Sand fraction	Gravel fraction
--------------	---------------	---------------	-----------------

On sieve	(% m/m) minimum	(% m/m) maximum
C 31.5	0	50
C 22.4	50	100
C 16.0	97	100
2 mm	98	100
Through sieve 63 μ m	≤ 1	

Crushed stone for crushed stone 2/6 lining**(Crushed stone 2/6)**

Sample Details			
Borehole No:	Sample No:	Sample Type:	Depth:



Lutum fract.	Silt fraction	Sand fraction	Gravel fraction

Other properties:

On sieve	(% m/m) minimum	(% m/m) maximum
C 8	-	0
C 5.6	0	5
C 4	15	45
2.8 mm	45	75
2.0 mm	80	100
1.4 mm	90	100
1 mm	95	100
Through sieve 63 μ m	≤ 1	

SAND BITUMEN

As a protective layer underneath tank bottom on top of sand, the sand bitumen mix consists of:

- 96 % (m/m) sand for tank foundation
- 4 % (m/m) bitumen 45/60

Application: the material shall be mixed warm and spread. Compaction shall be by means of a light roller.

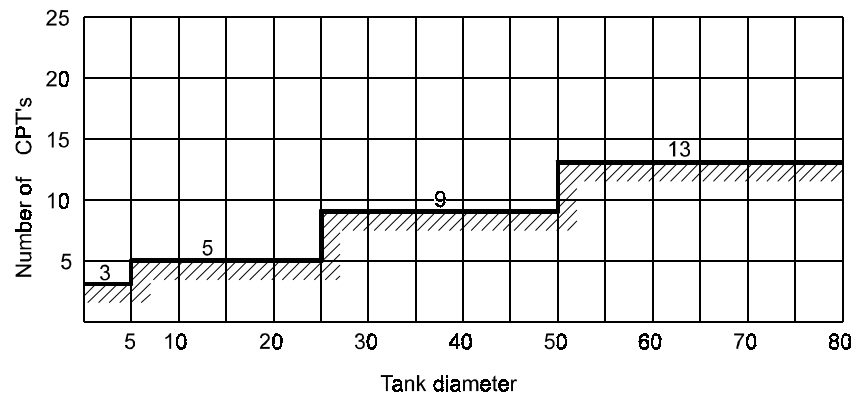
MICRO ASPHALT

This product can be used as a protective layer on the tank foundation shoulder and the slope of the tank foundation.

Micro asphalt consists of:

- | | |
|----------------------------|--------------|
| ■ broken sand | 47.6 % (m/m) |
| ■ crushed stone | 32.3 % (m/m) |
| ■ sand for tank foundation | 15.8 % (m/m) |
| ■ bitumen | 7.5 % (m/m) |
| ■ weak filling | 4.3 % (m/m) |

APPENDIX 14 NUMBER OF VERIFICATION CPTs FOR COMPLETED TANK FOUNDATION



APPENDIX 15 TYPICAL MINIMUM CONE RESISTANCES FOR SOIL COMPACTION

